



Swiss Institute of
Bioinformatics

Introduction to RNA-Seq: Overview

Wandrille Duchemin

General Information

Course page: <https://sib-swiss.github.io/RNAseq-introduction-training/>

- Slides, Data sets, Exercises, Solutions



Optional exam, 0.5 ECTS value

- Course from 09:00 to 17:00
- Lunch break 12:00 to 13:00
- 15min breaks around 10:30 and 15:00

Asking questions - Communication

- Raise your hand anytime



- Done with an exercise?



Course Outline

Day 1

1. **Overview** of RNAseq
2. Getting started with the **cluster**
3. **Quality Control** of the raw data
4. Sequence **trimming**

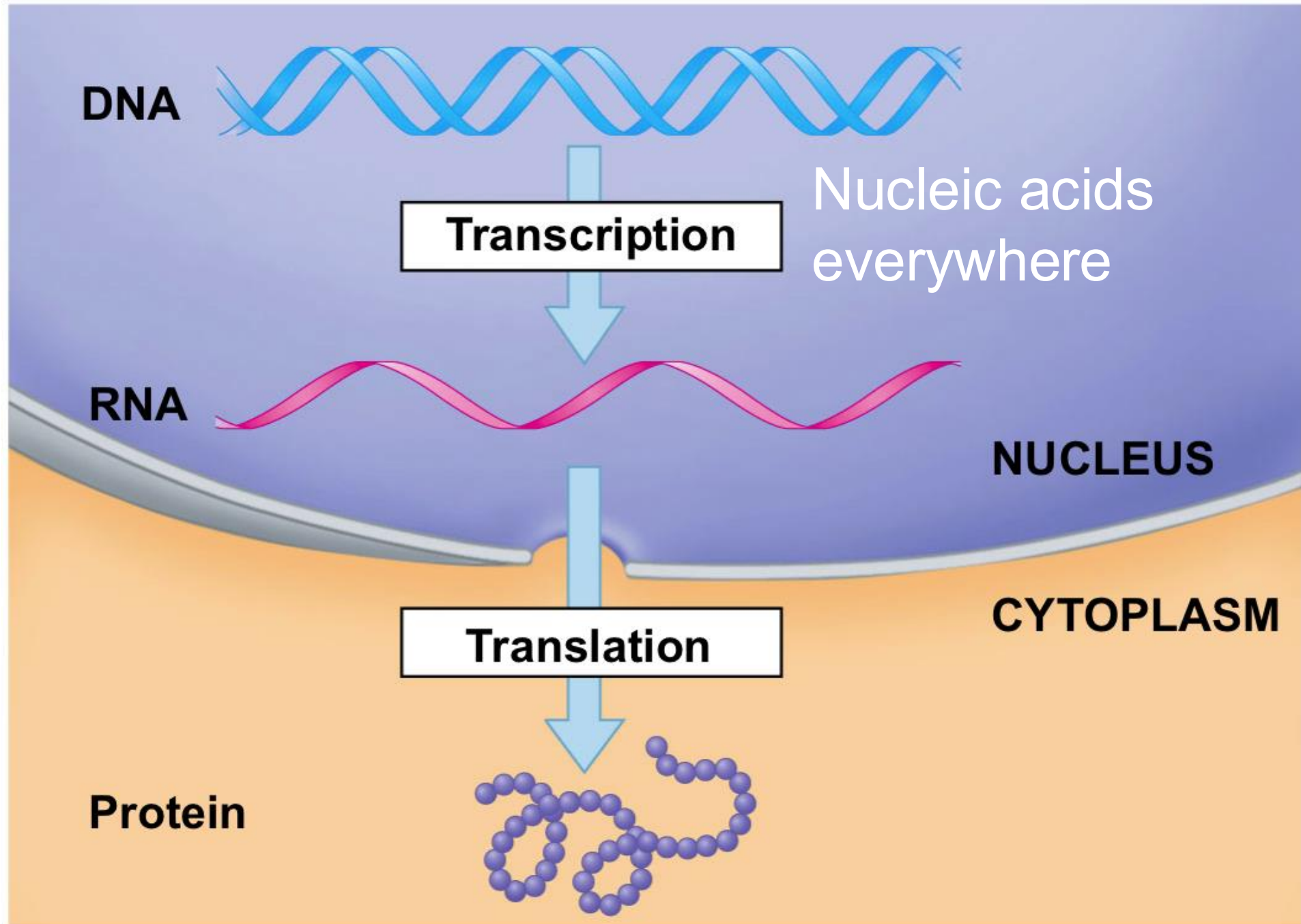
Day 2

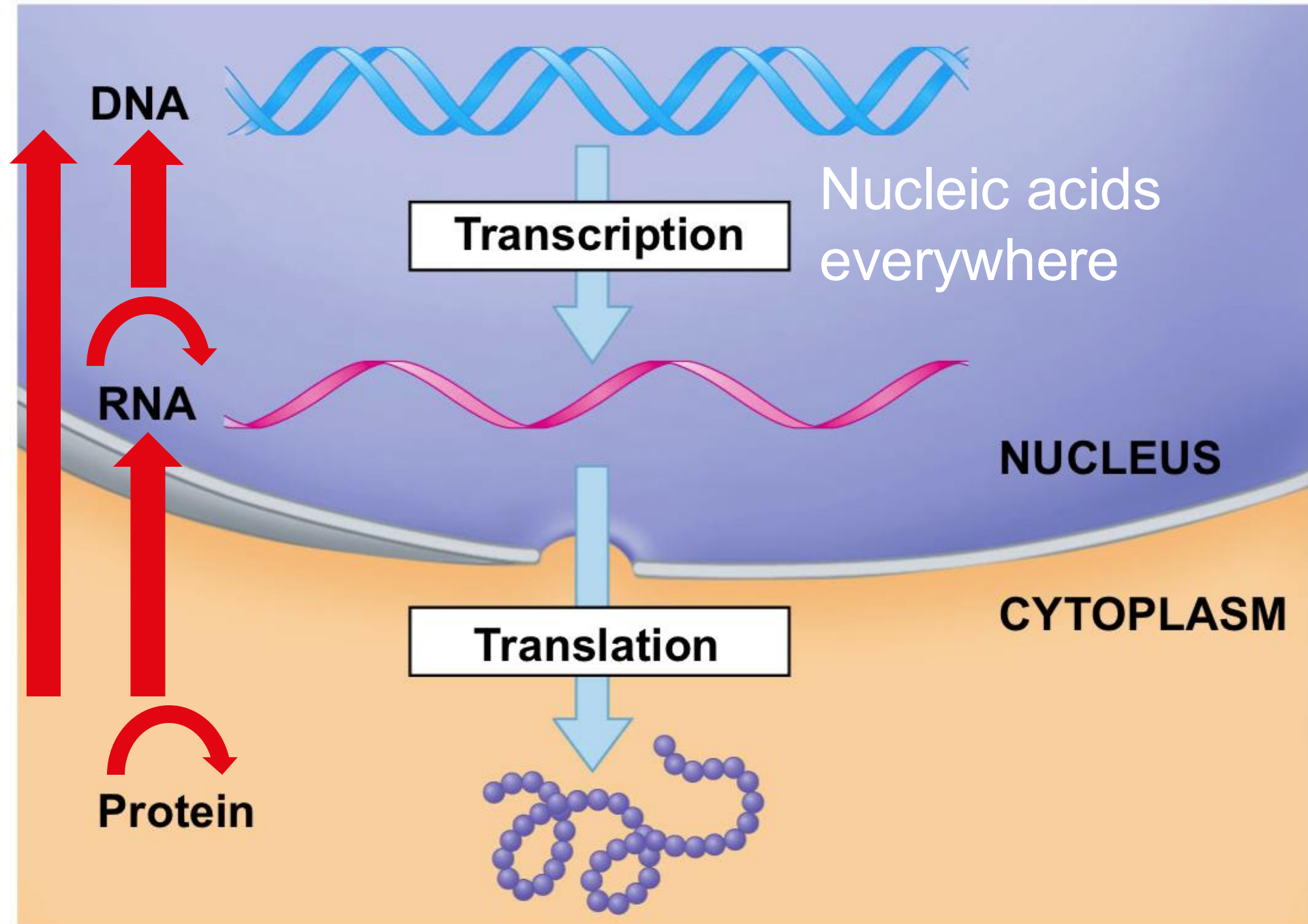
1. Reas **mapping**
2. **Differential Expression** Inference
3. Enrichment Analysis

slides Outline

- RNA and molecular biology
- Main challenges for RNAseq
- Major Sequencing technologies
- Planning your sequencing : choices, number of samples, ...
- Bioinformatics analysis overview

Introducing Ourselves





alternative splicing adds a layer of complexity

~20'000 mammalian genes

>>100'000 (?) transcripts

>>1'000'000 (?) proteins

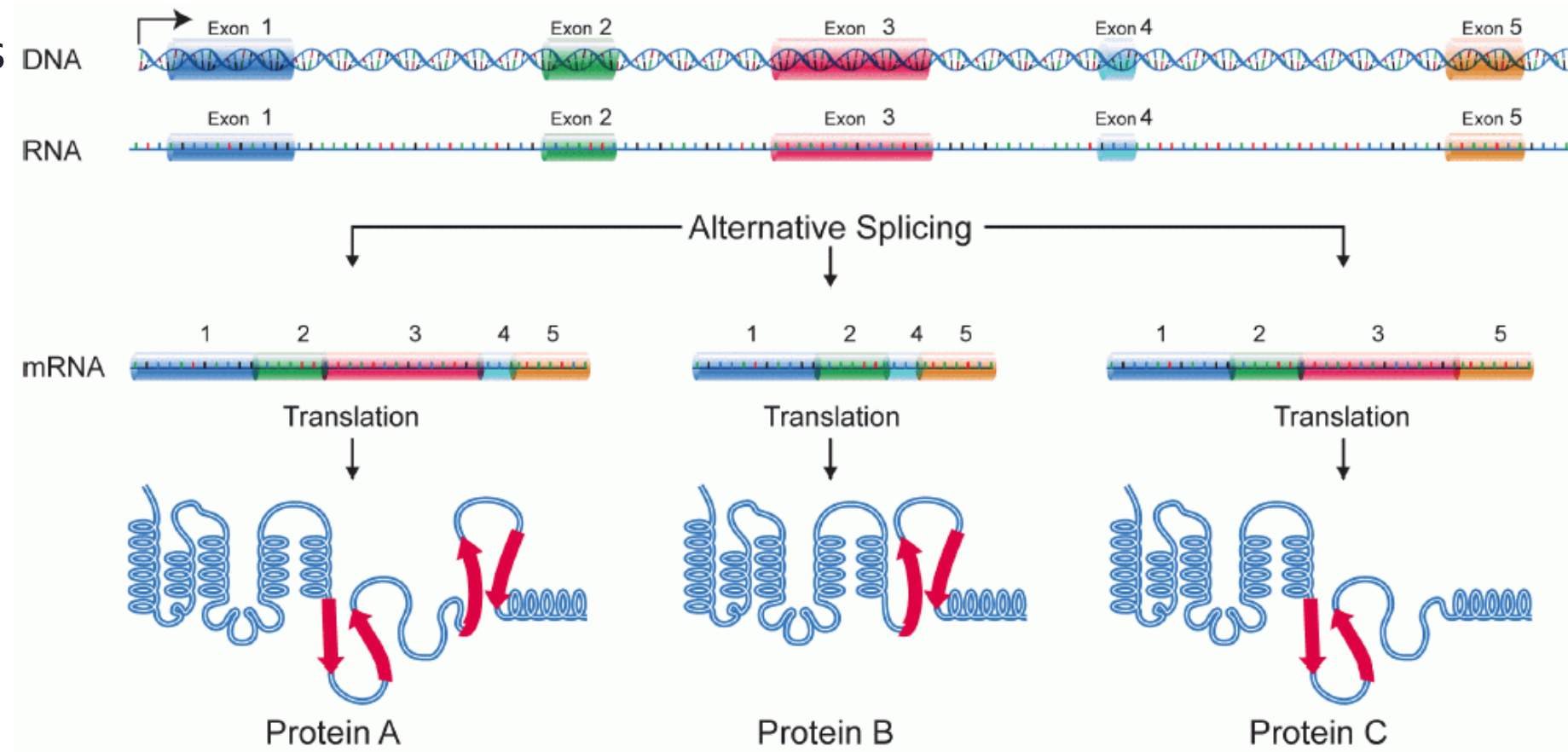
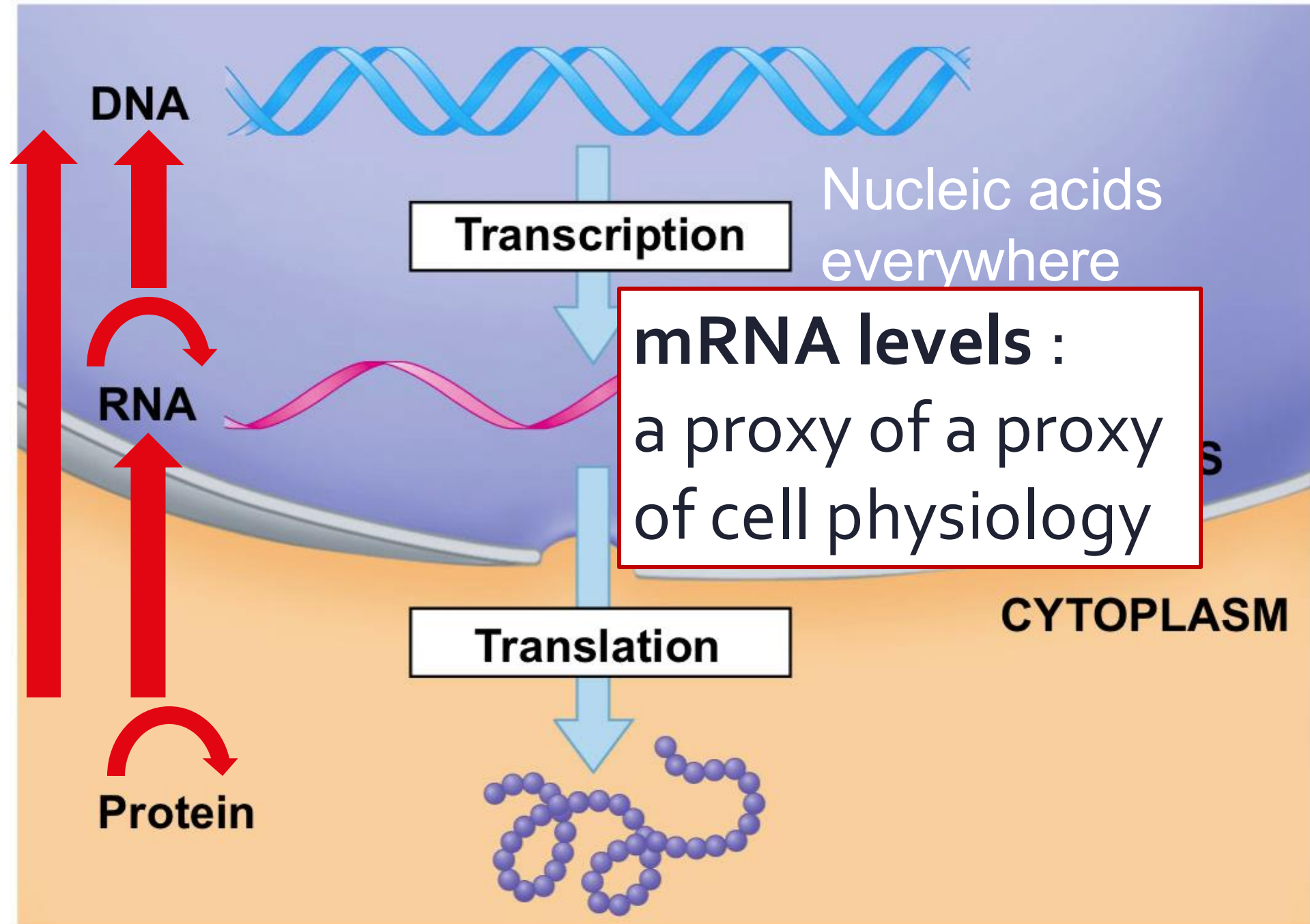


Image credit: National Human Genome Research Institute – public domain



What (and why) are we sequencing

Genomics

- Whole genome/exome sequencing (WGS/WES)
- Variant calling (SNPs, CNVs, structural variations)

Epigenomics

- Bisulphite sequencing : DNA methylation
- ATAC-Seq : chromatine opening
- ChIP-seq : TF binding sites

Transcriptomics

- Total RNA
- Poly-A tail selection : focus on mRNA
- Ribo depletion: mRNA + ncRNA
- 5'/3' RACE seq : isoform characterization for one gene
- scRNAseq
- Long read RNA sequencing
- ...

What (and why) are we sequencing

Genomics

- Whole genome/exome sequencing (WGS/WES)
- Variant calling (SNPs, CNVs, structural variations)

Imagination is the limit

Epigenomics

- Bisulphite sequencing : DNA methylation
- ATAC-Seq : chromatine opening
- ChIP-seq : TF binding sites

Transcriptomics

- Total RNA
- Poly-A tail selection : focus on mRNA
- Ribo depletion: mRNA + ncRNA
- 5'/3' RACE seq : isoform characterization for one gene
- scRNAseq
- Long read RNA sequencing
- ...

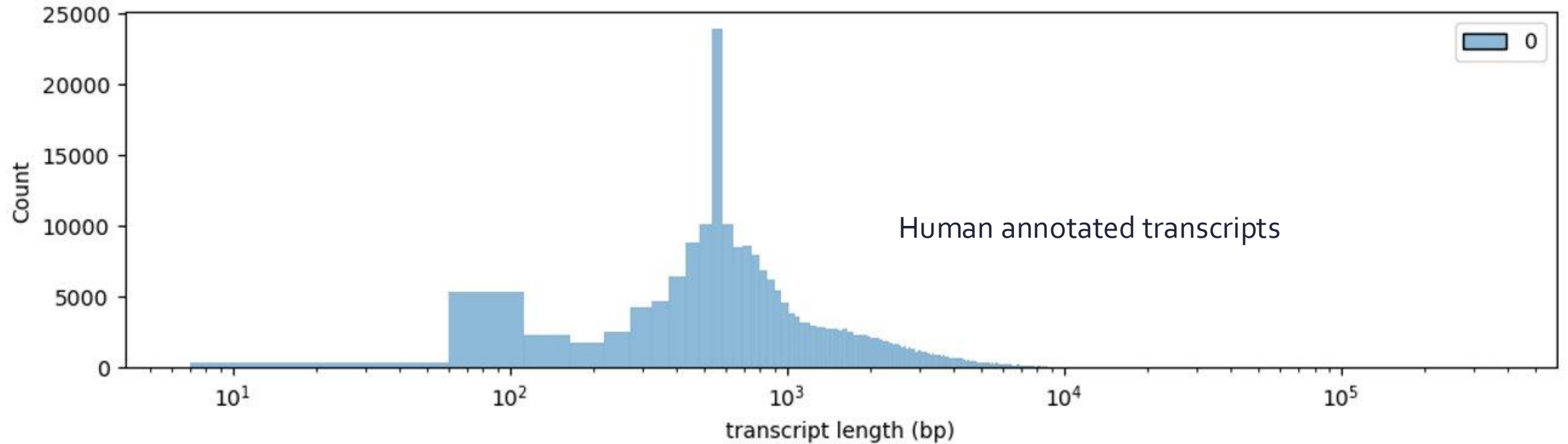
See : <https://liorpachter.wordpress.com/seq/>

slides Outline

- RNA and molecular biology
- **Main challenges for RNAseq**
- Major Sequencing technologies
- Planning your sequencing : choices, number of samples, ...
- Bioinformatics analysis overview

Main challenges of RNAseq

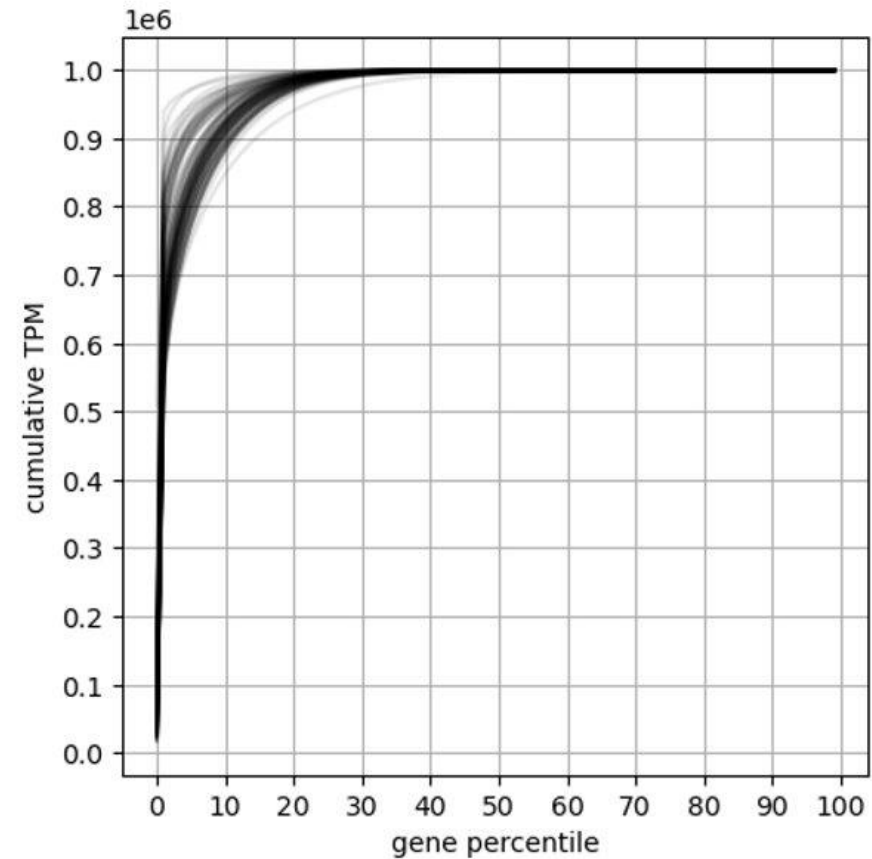
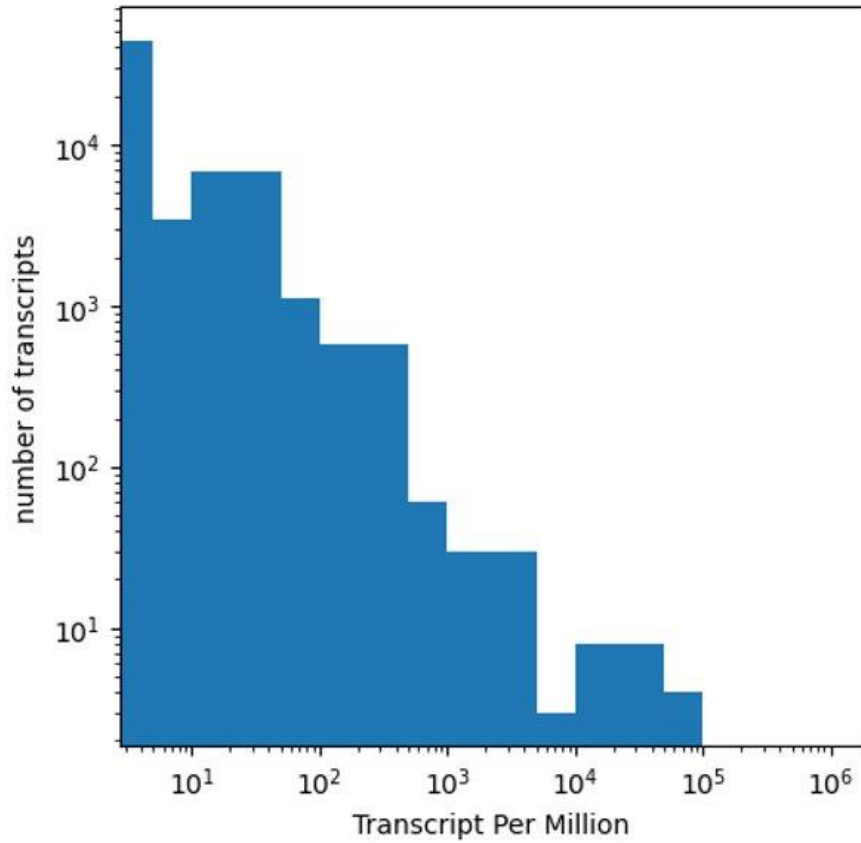
Transcripts are diverse in size



Main challenges of RNAseq

Transcripts are diverse in size

Expression levels have a *high dynamic range*



From Gtex V8 – human tissue samples

Data source : https://gtexportal.org/home/downloads/adult-gtex/bulk_tissue_expression

Main challenges of RNAseq

Transcripts are diverse in size

Expression levels have a *high dynamic range*

RNA molecules are exposed to degradation enzyme:

- RNA integrity affects results

Is there a reference genome.

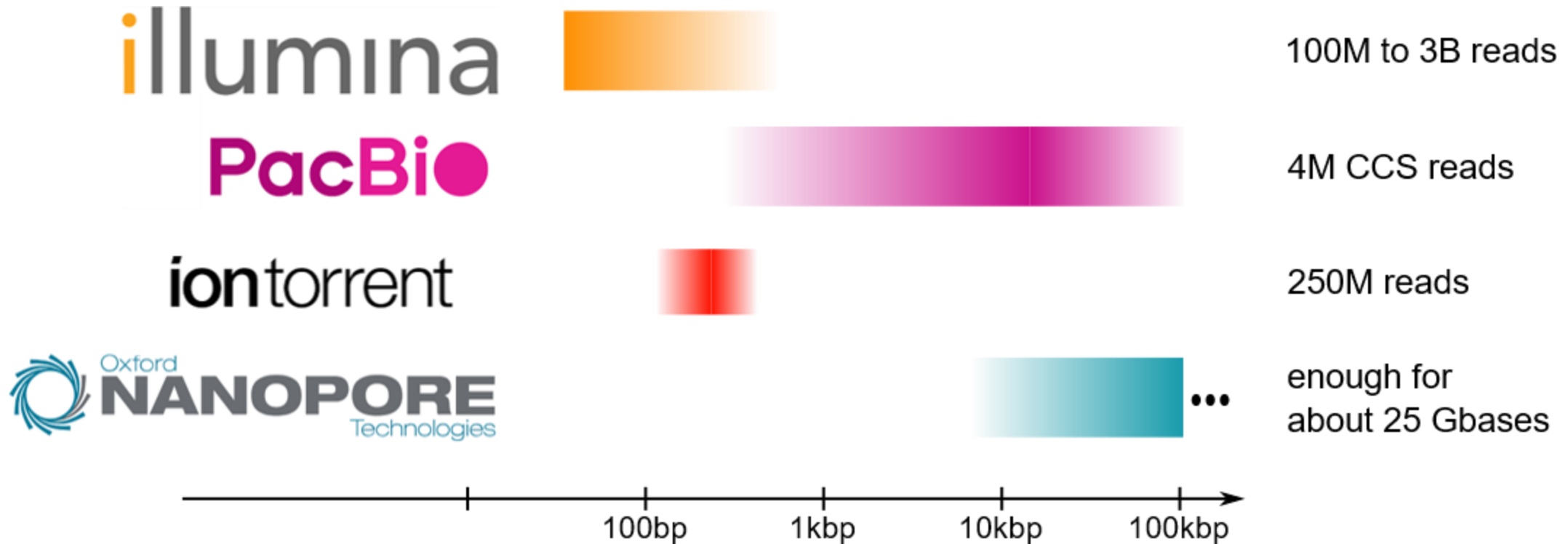
If yes,

- How good is it?
- How good is the gene annotation ?

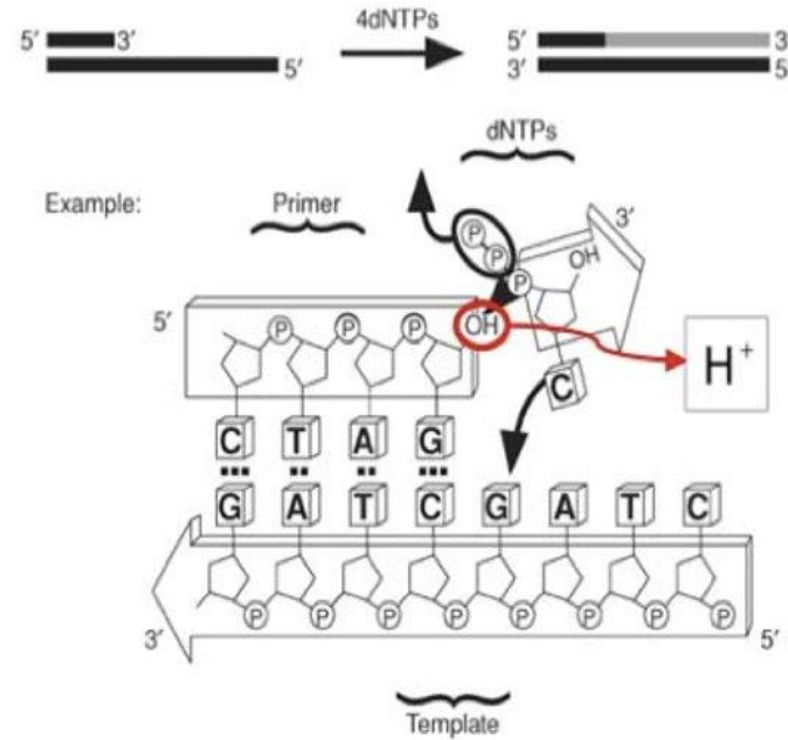
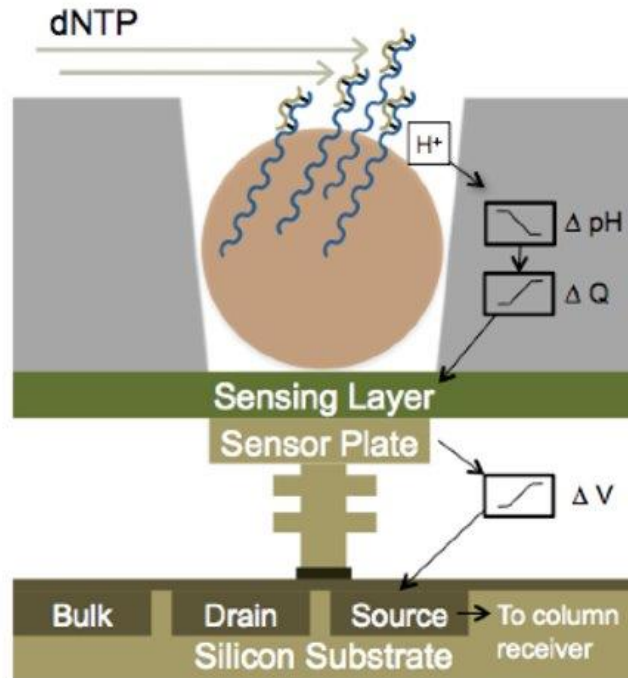
slides Outline

- RNA and molecular biology
- Main challenges for RNAseq
- **Major Sequencing technologies**
- Planning your sequencing : choices, number of samples, ...
- Bioinformatics analysis overview

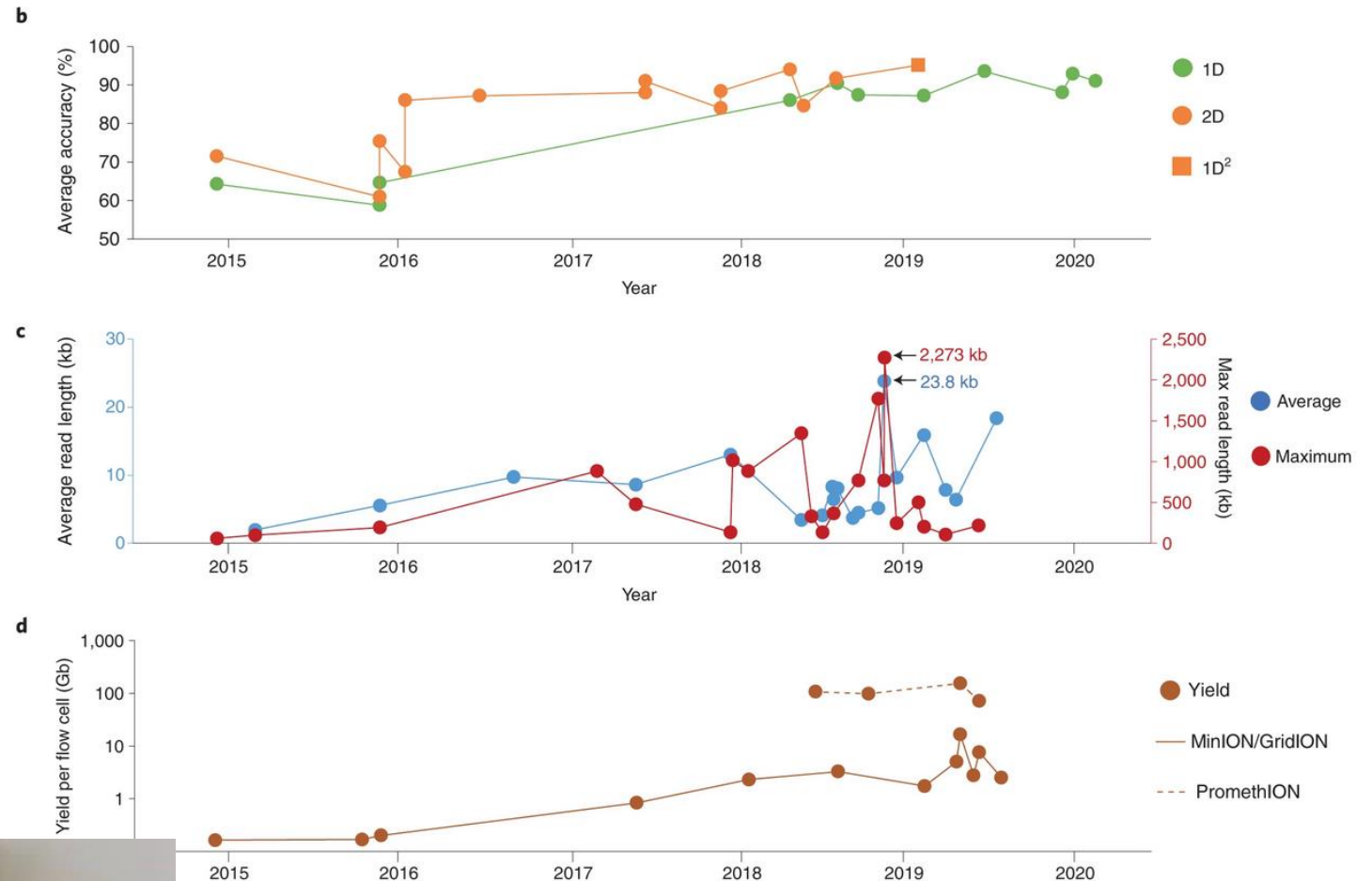
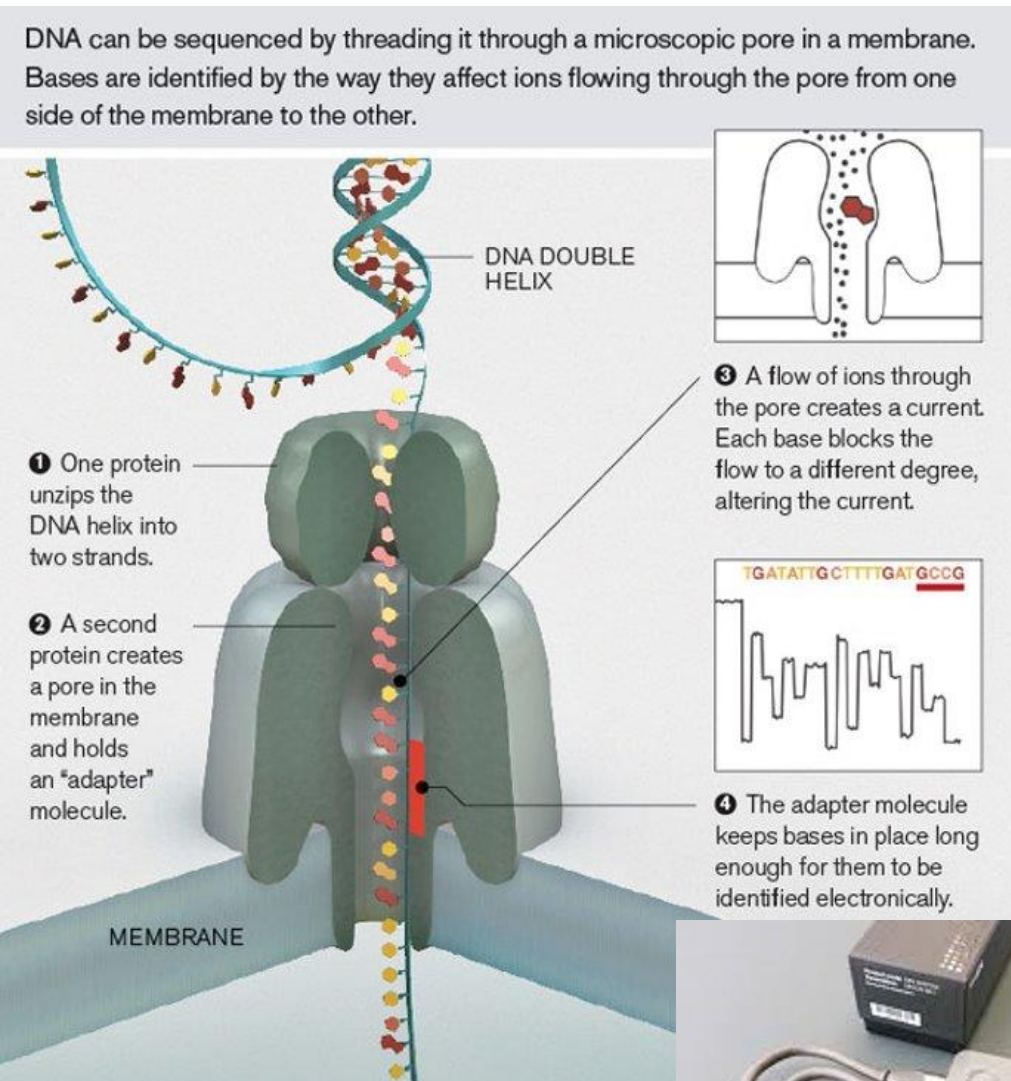
Main sequencing technologies



Ion torrent - reading pH changes



Oxford Nanopore - direct sequencing



From Wang, Y., *et al.* Nanopore sequencing technology, bioinformatics and applications. *Nat Biotechnol* **39**, 1348–1365 (2021). <https://doi.org/10.1038/s41587-021-01108-x>

Pacific Biosciences - Single Molecule Real Time

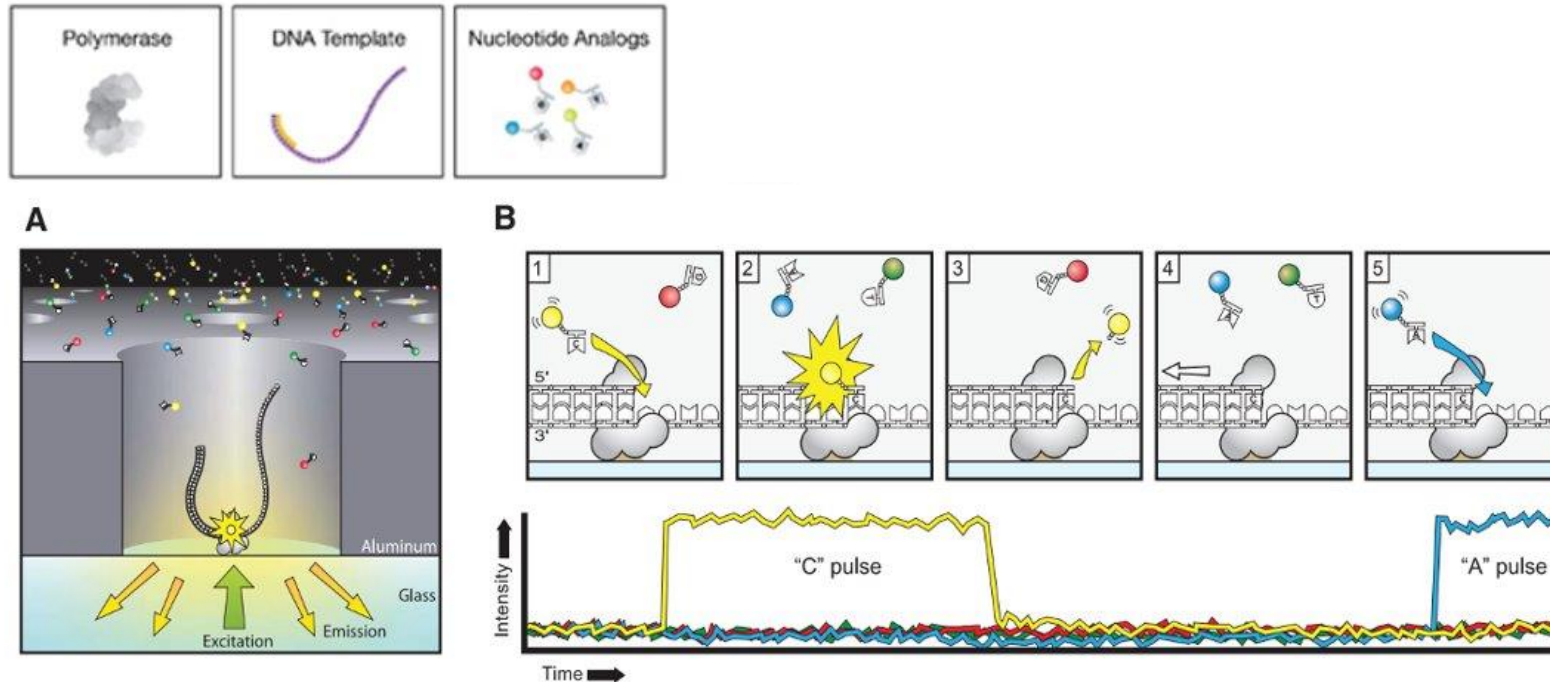


Fig. 1. Principle of single-molecule, real-time DNA sequencing. **(A)** Experimental geometry. A single molecule of DNA template-bound $\Phi 29$ DNA polymerase is immobilized at the bottom of a ZMW, which is illuminated from below by laser light. The ZMW nanostructure provides excitation confinement in the zeptoliter (10^{-21} liter) regime, enabling detection of individual phospholinked nucleotide substrates against the bulk solution background as they are incorporated into the DNA strand by the polymerase. **(B)** Schematic event sequence of the phospholinked dNTP incorporation cycle,

with a corresponding expected time trace of detected fluorescence intensity from the ZMW. (1) A phospholinked nucleotide forms a cognate association with the template in the polymerase active site, (2) causing an elevation of the fluorescence output on the corresponding color channel. (3) Phosphodiester bond formation liberates the dye-linker-pyrophosphate product, which diffuses out of the ZMW, thus ending the fluorescence pulse. (4) The polymerase translocates to the next position, and (5) the next cognate nucleotide binds the active site beginning the subsequent pulse.

Pacific Biosciences - Circular Consensus Sequencing

Raw reads
~15% random errors

Start with high-quality
double stranded DNA



Ligate SMRTbell
adapters and size select



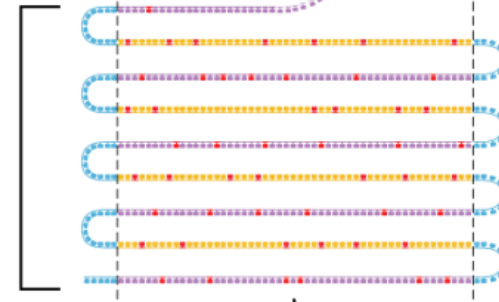
Anneal primers and
bind DNA polymerase



Circularized DNA
is sequenced in
repeated passes

The polymerase reads
are trimmed of adapters
to yield subreads

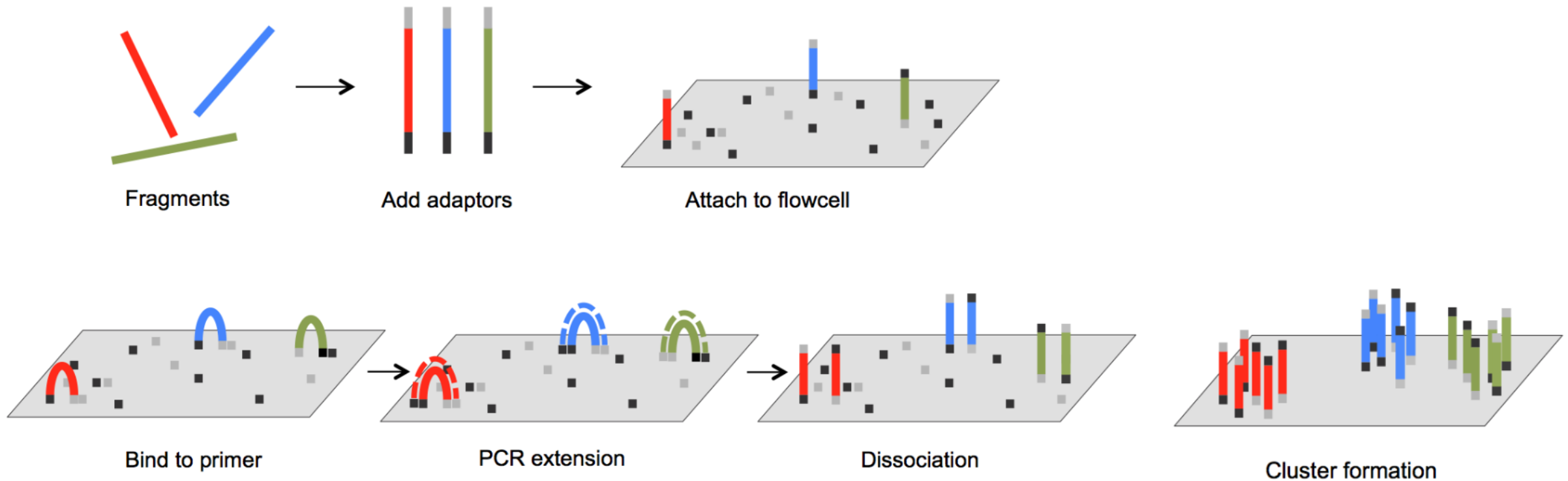
Consensus is called
from subreads



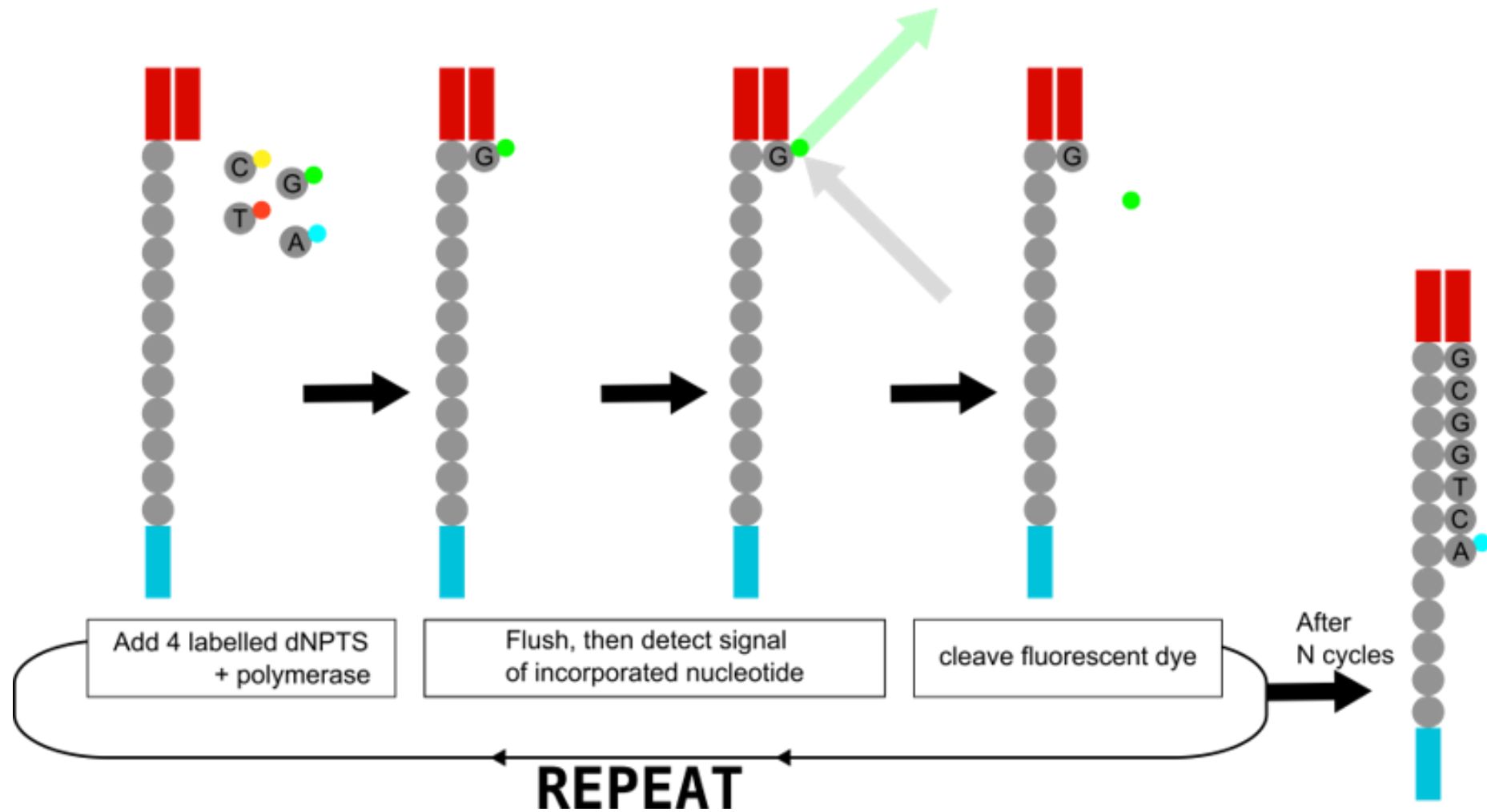
HiFi READ
(>99% accuracy)

Typical in isoseq

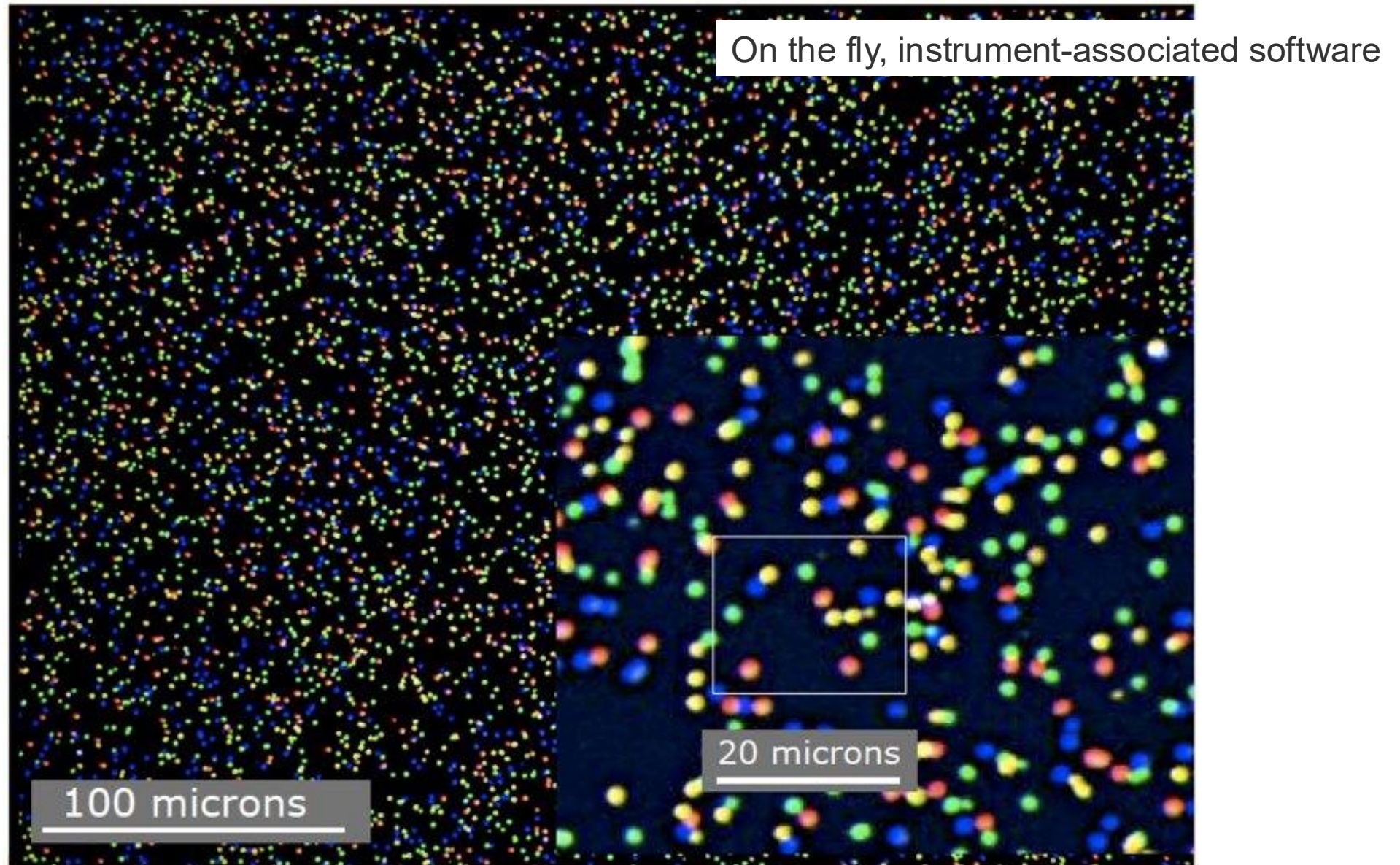
Illumina sequencing - cluster formation



Illumina sequencing - sequencing by synthesis

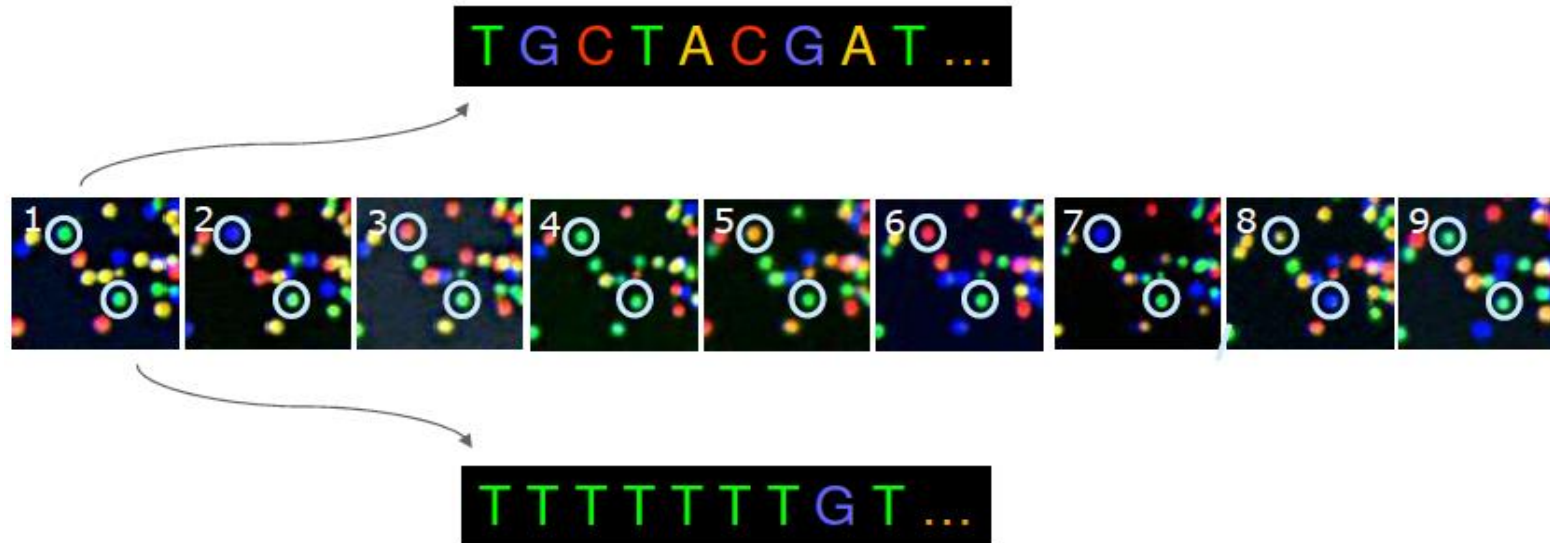


Illumina sequencing - image analysis



Illumina sequencing - from image to sequence

Base Calling From Raw Data



The identity of each base of a cluster is read off from sequential images

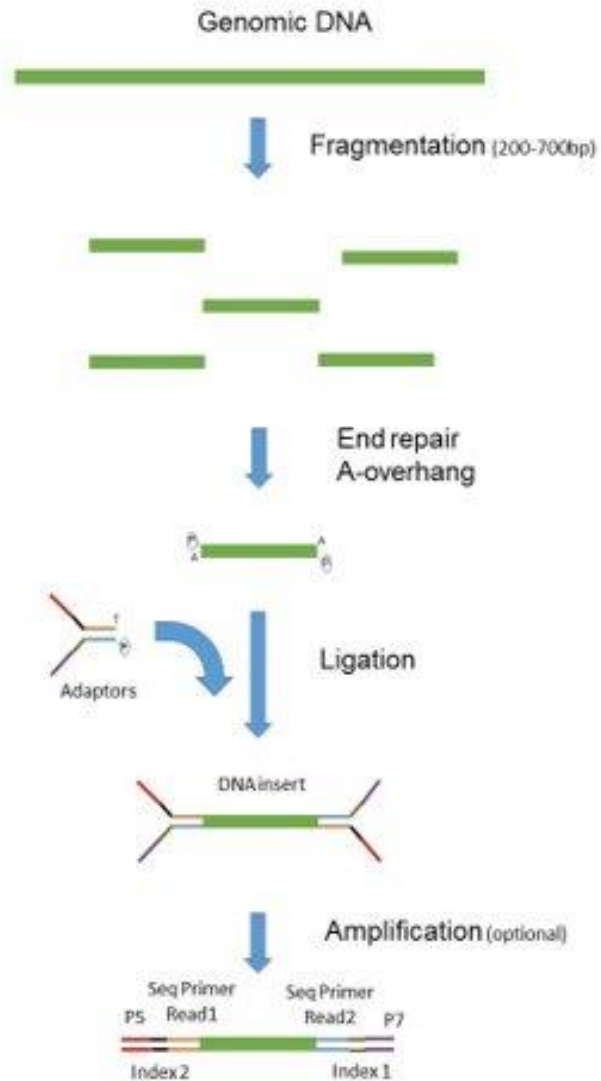


slides Outline

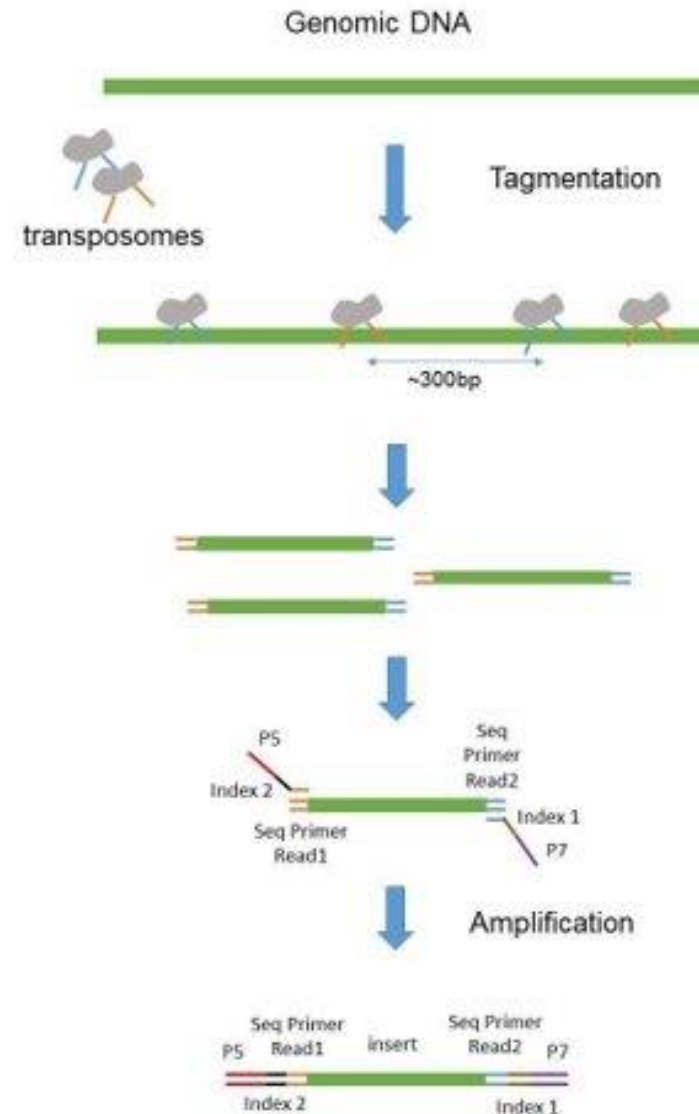
- RNA and molecular biology
- Main challenges for RNAseq
- Major Sequencing technologies
- **Planning your sequencing : choices, number of samples, ...**
- Bioinformatics analysis overview

Paired-end sequencing

“Classical” paired end library (illumina)

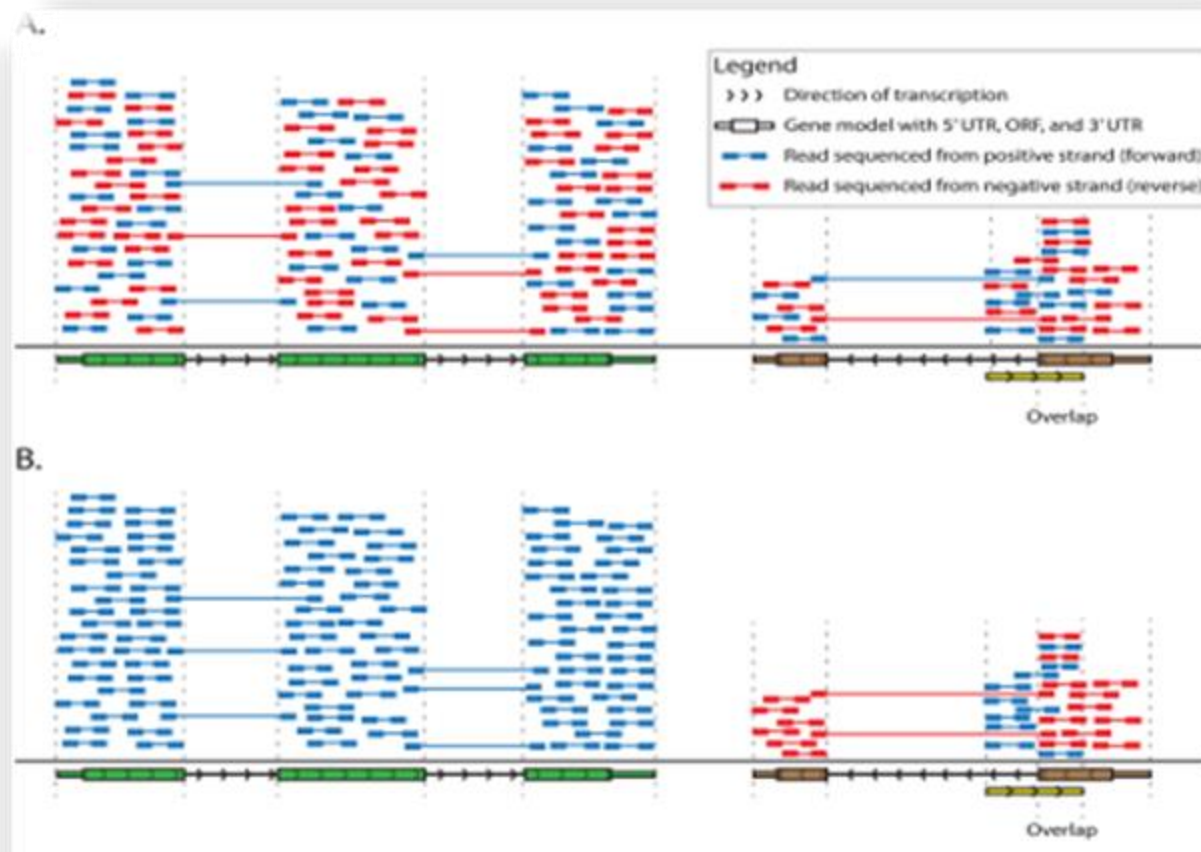


“Nextera” paired end library



Stranded vs Unstranded Sequencing

- Overlapping genes regions are substantial (~8% in *Homo sapiens*)
- Stranded sequencing allows us to quantify expression in these overlapping regions
- Achieved by ligating different adapters to 5' and 3' ends



RNA purification

PolyA selection

- Commonly used and inexpensive
- 3' end bias when RNA is degraded
- Loses almost all non-polyA transcripts
- Gets rid of vast majority of ribosomal RNAs, but ncRNA too

Ribosomal RNA depletion

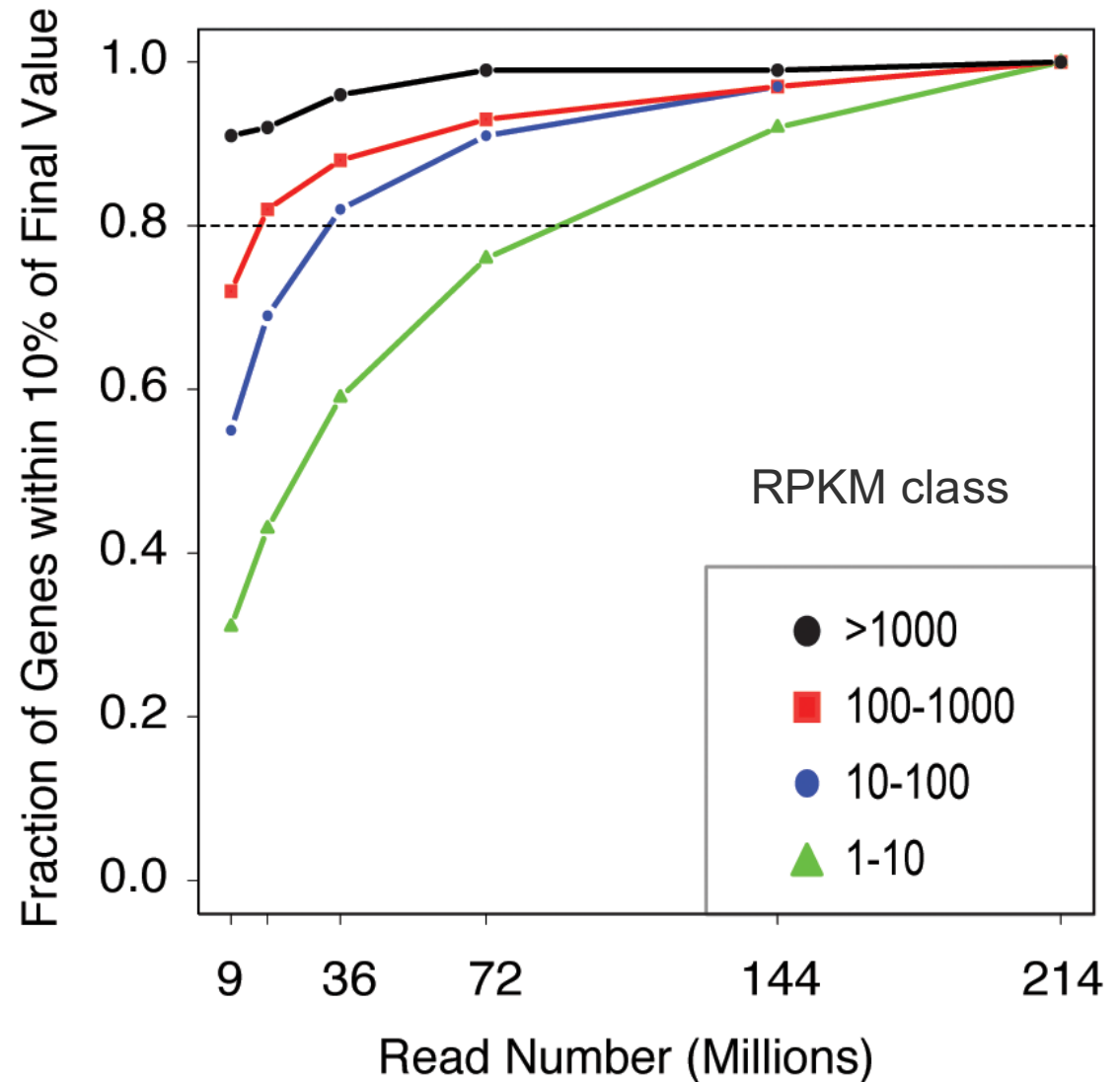
- Less popular, ~2x more expensive
- Higher proportion of rRNA than in polyA selection
- Bacterial data
- Allows identification of lncRNAs without polyA tails
- Retains more immature mRNAs (bad for gene expression quantification)

Sequencing depth

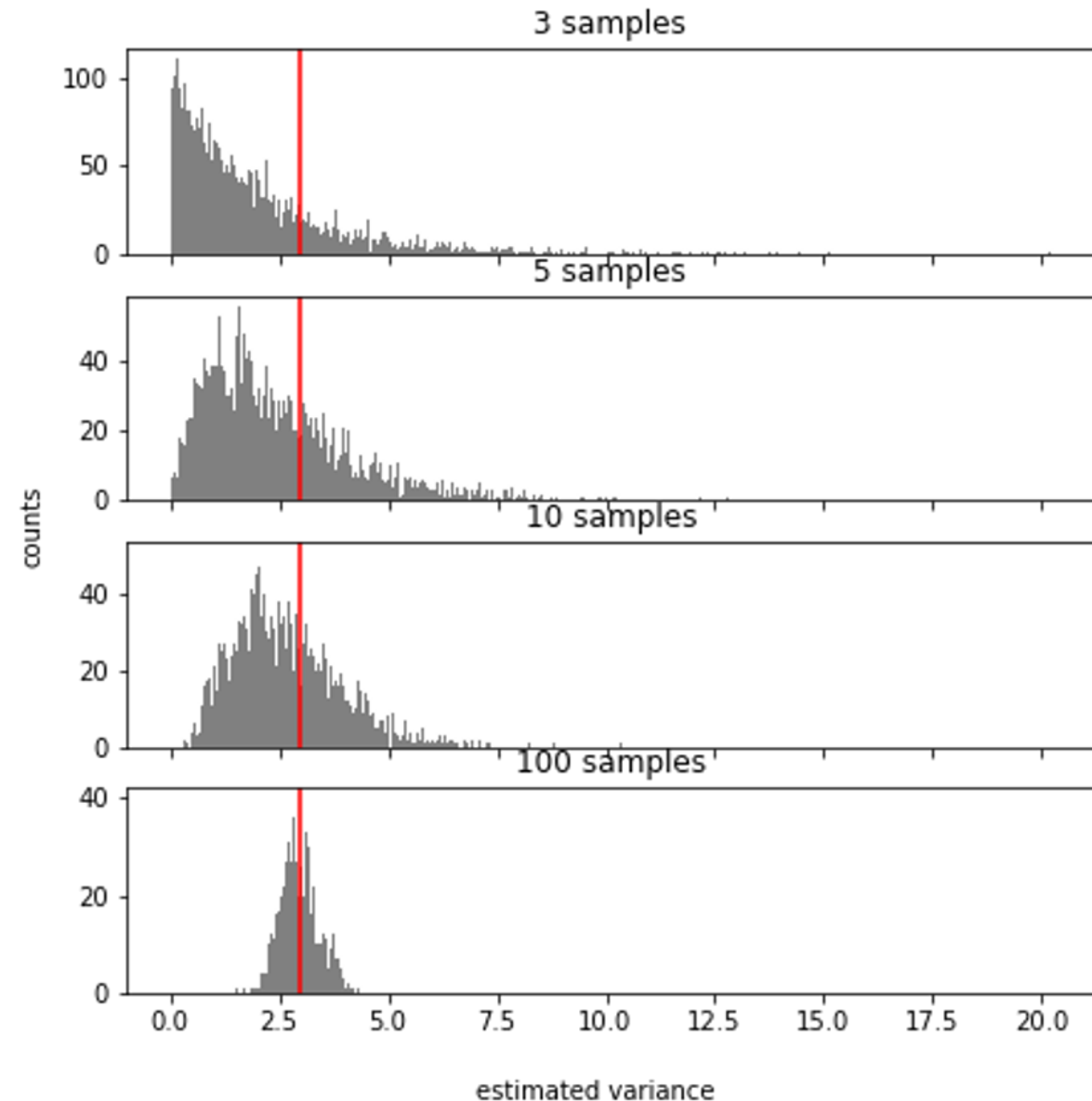
DE : usually aim for ~30-40 million reads

For rare events (isoforms, somatic mutations)
much more depth is required

Not easy to know in advance

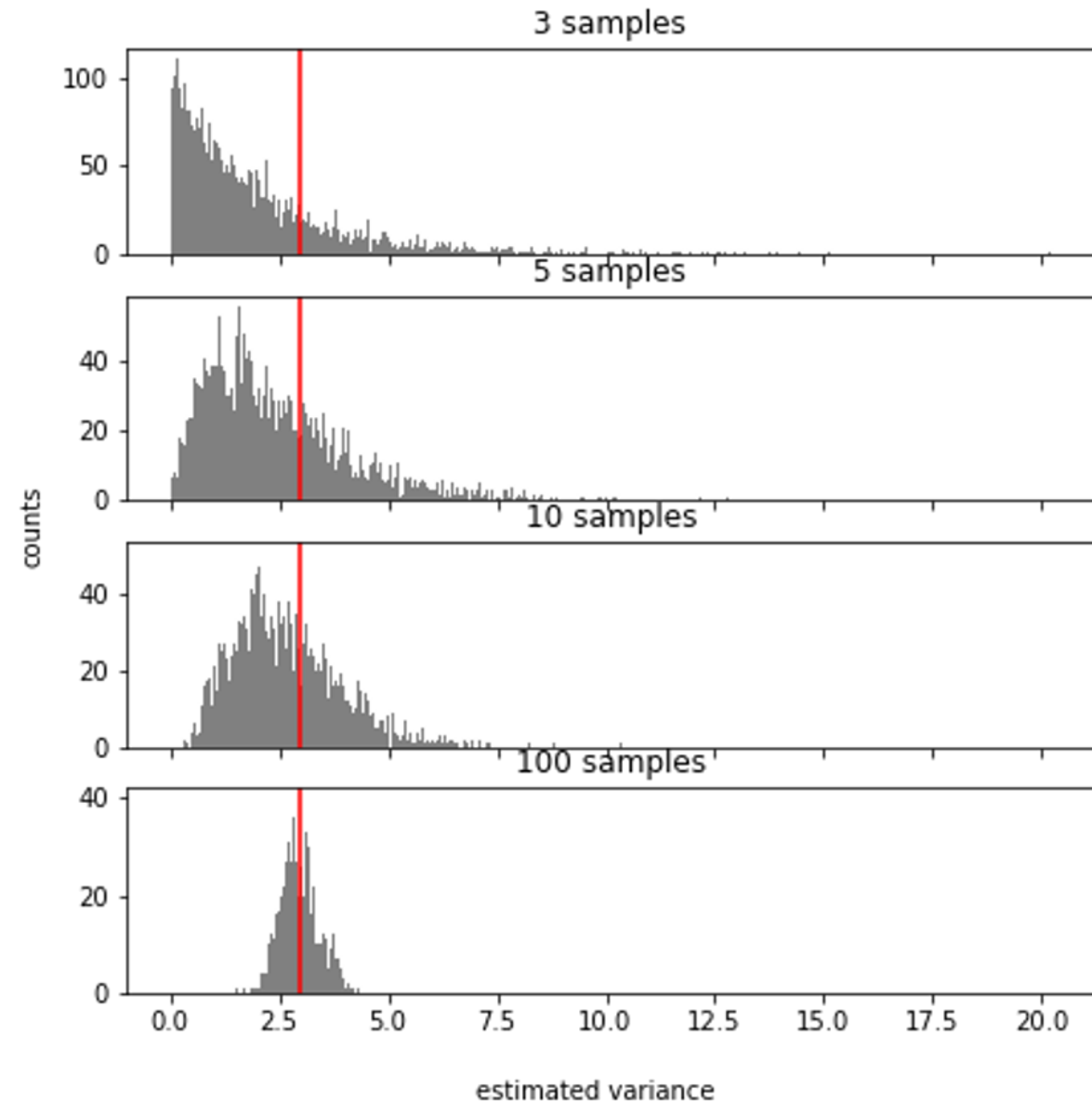


Replicates - estimating a biological variance



What does this tell you about the number of replicates needed?

Replicates - estimating a biological variance

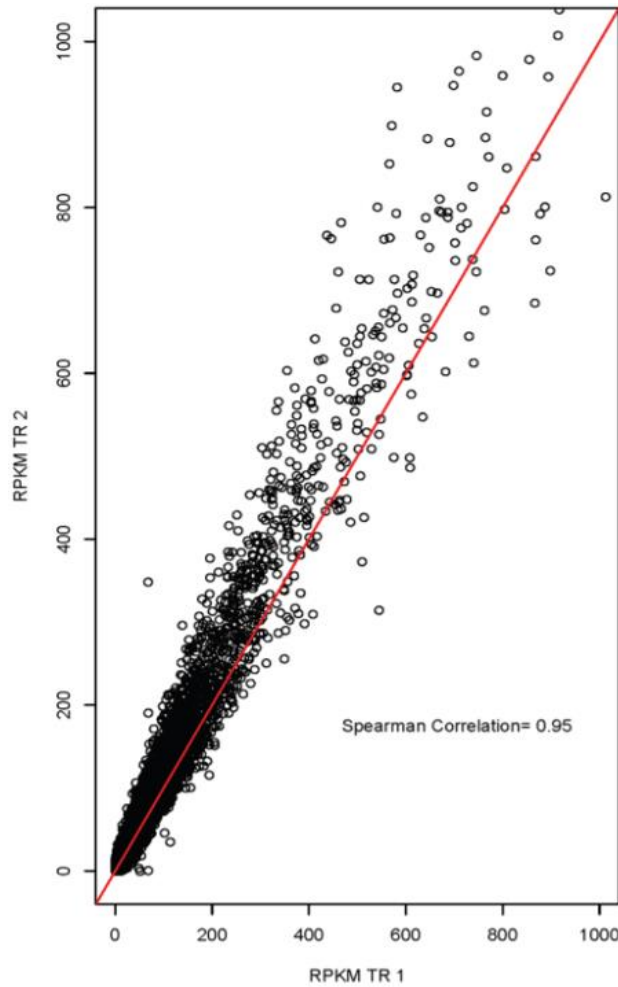


What does this tell you about the number of replicates needed?

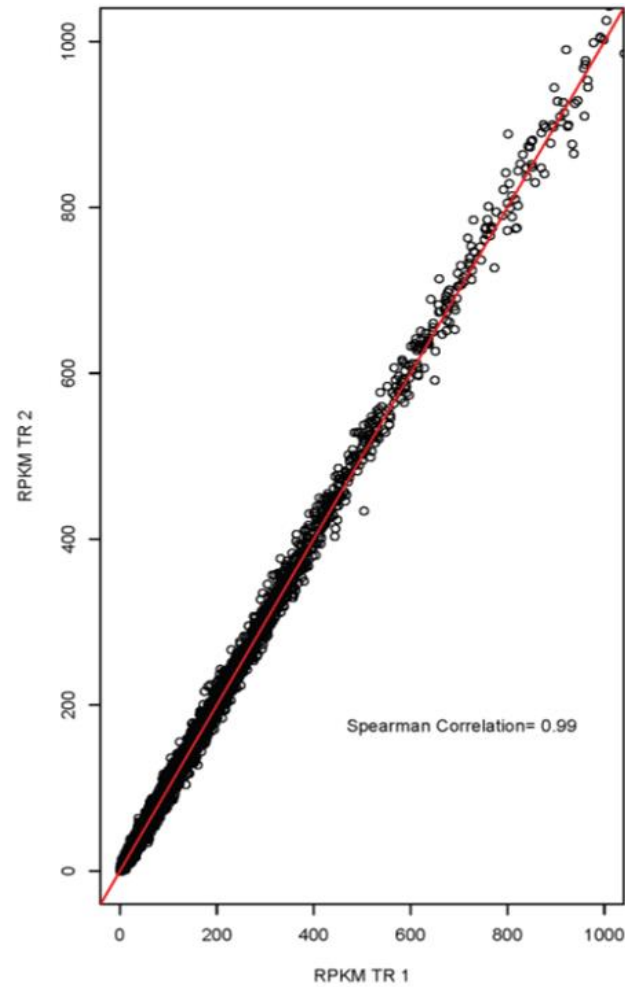
2 types of replicates:

- **Technical:** same RNA extract
- **Biological:** same biological condition

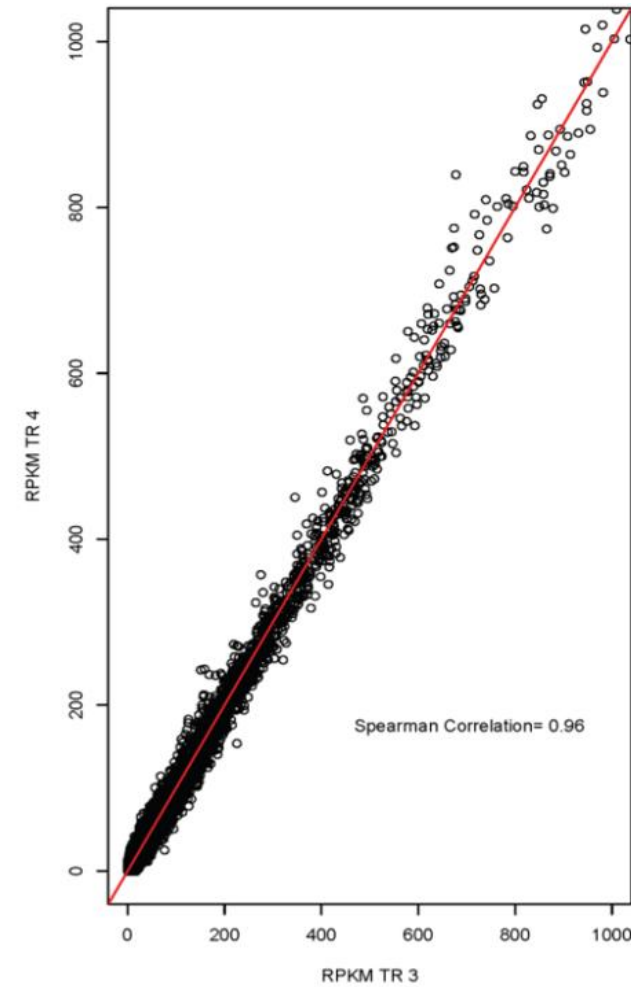
Technical replicates



D. simulans
Male heads



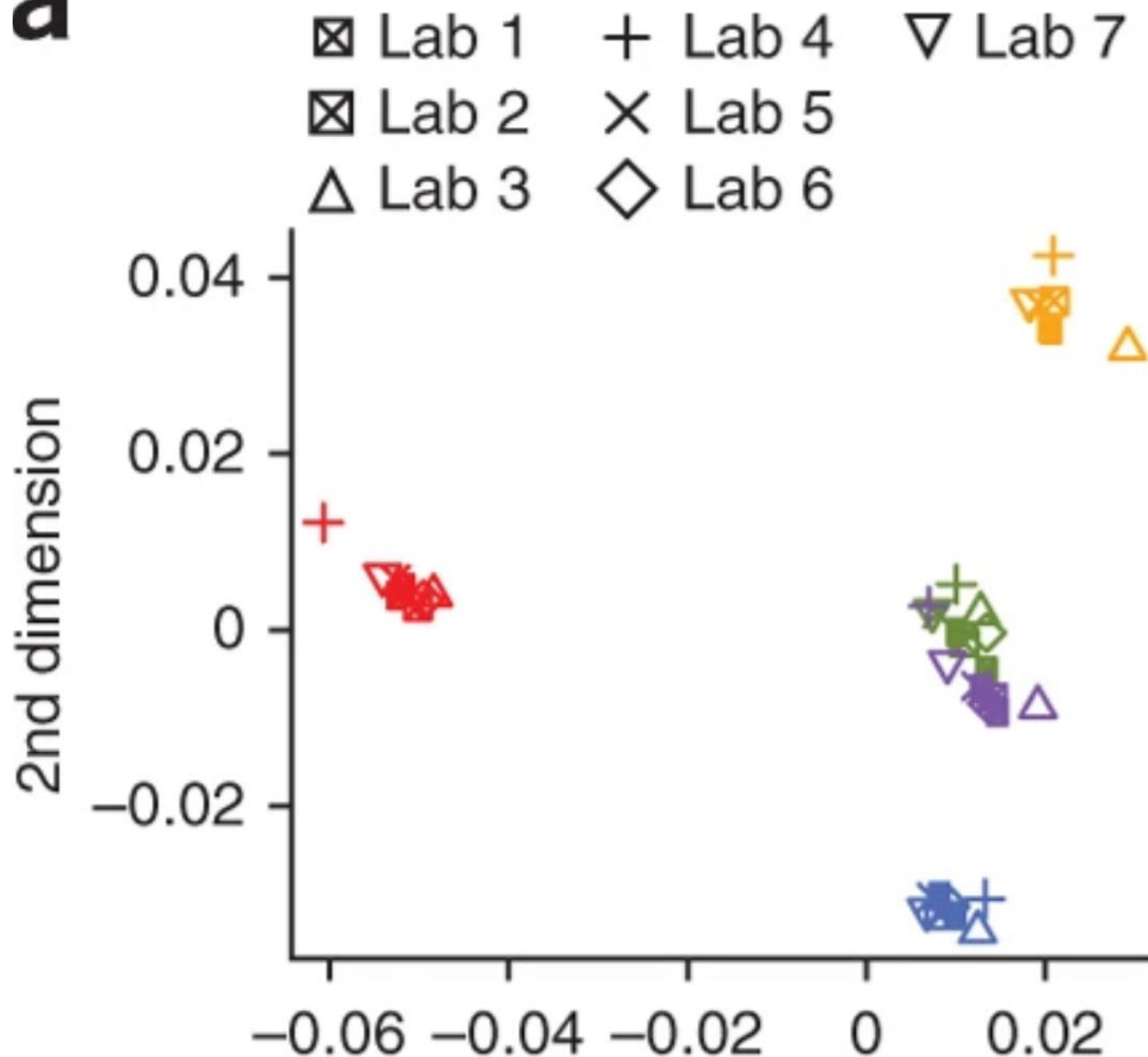
D. melanogaster
Female heads



C167 cell line

Technical replicates

a



Article | Published: 01 November 2013

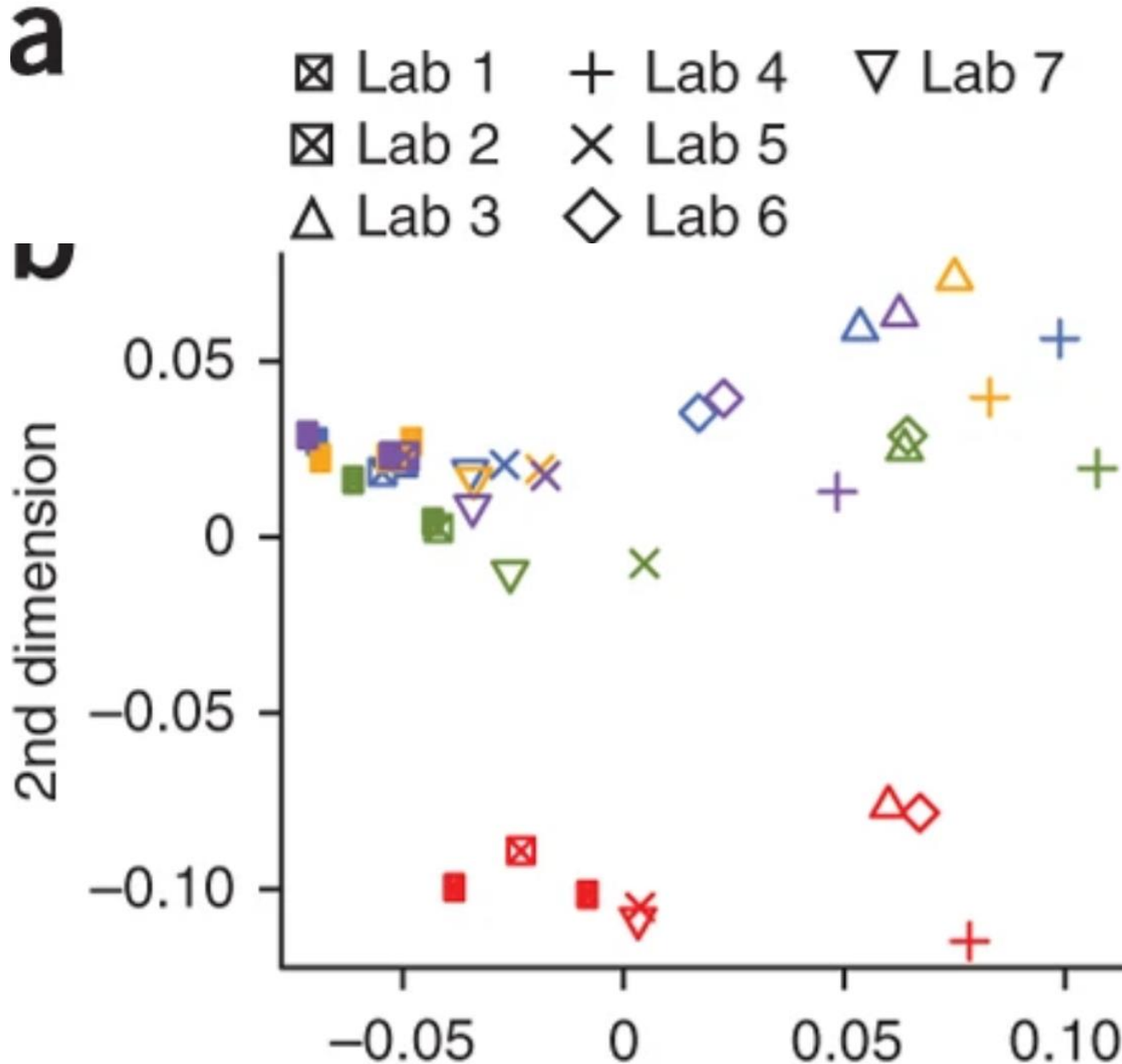
Reproducibility of high-throughput mRNA and small RNA sequencing across laboratories

[Peter A C 't Hoen](#) , [Marc R Friedländer](#), [Jonas Almlöf](#), [Michael Sammeth](#), [Irina Pulyakhina](#), [Seyed Yahya Anvar](#), [Jeroen F J Laros](#), [Henk P J Buermans](#), [Olof Karlberg](#), [Mathias Brännvall](#), [The GEUVADIS Consortium](#), [Johan T den Dunnen](#), [Gert-Jan B van Ommen](#), [Ivo G Gut](#), [Roderic Guigó](#), [Xavier Estivill](#), [Ann-Christine Syvänen](#), [Emmanouil T Dermitzakis](#) & [Tuuli Lappalainen](#) 

Nature Biotechnology **31**, 1015–1022 (2013) | [Cite this article](#)

Exon level: reproducible

Technical replicates



Article | Published: 01 November 2013

Reproducibility of high-throughput mRNA and small RNA sequencing across laboratories

[Peter A C 't Hoen](#) , [Marc R Friedländer](#), [Jonas Almlöf](#), [Michael Sammeth](#), [Irina Pulyakhina](#), [Seyed Yahya Anvar](#), [Jeroen F J Laros](#), [Henk P J Buermans](#), [Olof Karlberg](#), [Mathias Brännvall](#), [The GEUVADIS Consortium](#), [Johan T den Dunnen](#), [Gert-Jan B van Ommen](#), [Ivo G Gut](#), [Roderic Guigó](#), [Xavier Estivill](#), [Ann-Christine Syvänen](#), [Emmanouil T Dermitzakis](#) & [Tuuli Lappalainen](#) 

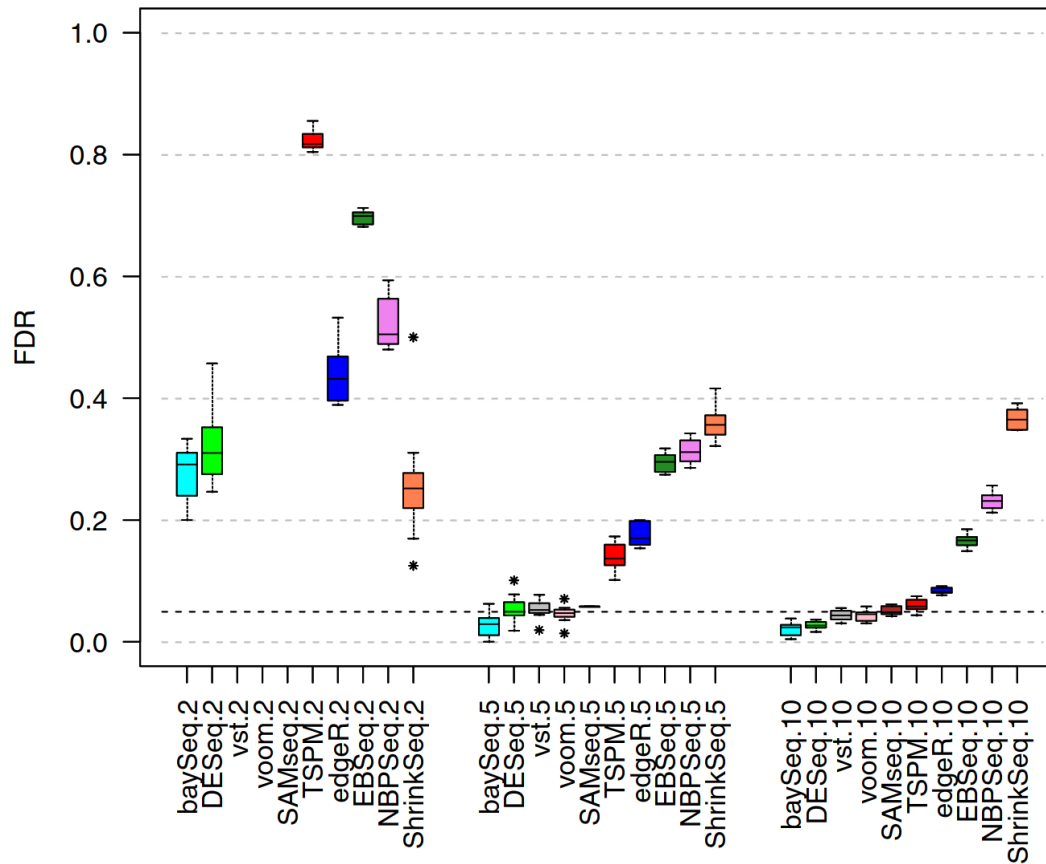
Nature Biotechnology **31**, 1015–1022 (2013) | [Cite this article](#)

Transcript level: not reproducible

Biological replicates

B

True FDR at $p_{\text{adj}} < 0.05$, B_{625}^{625}



2

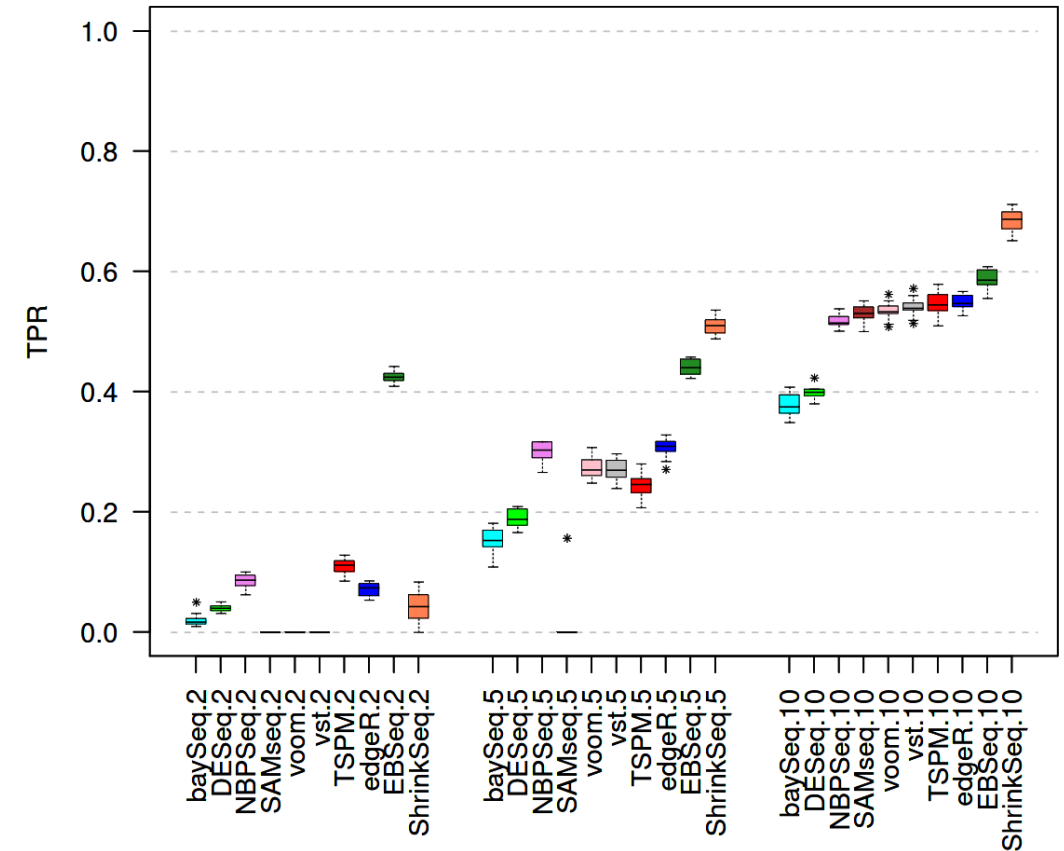
5

10

Samples per condition

Soneson, C., Delorenzi, M. A comparison of methods for differential expression analysis of RNA-seq data. *BMC Bioinformatics* **14**, 91 (2013).

TPR at $p_{\text{adj}} < 0.05$, B_{625}^{625}

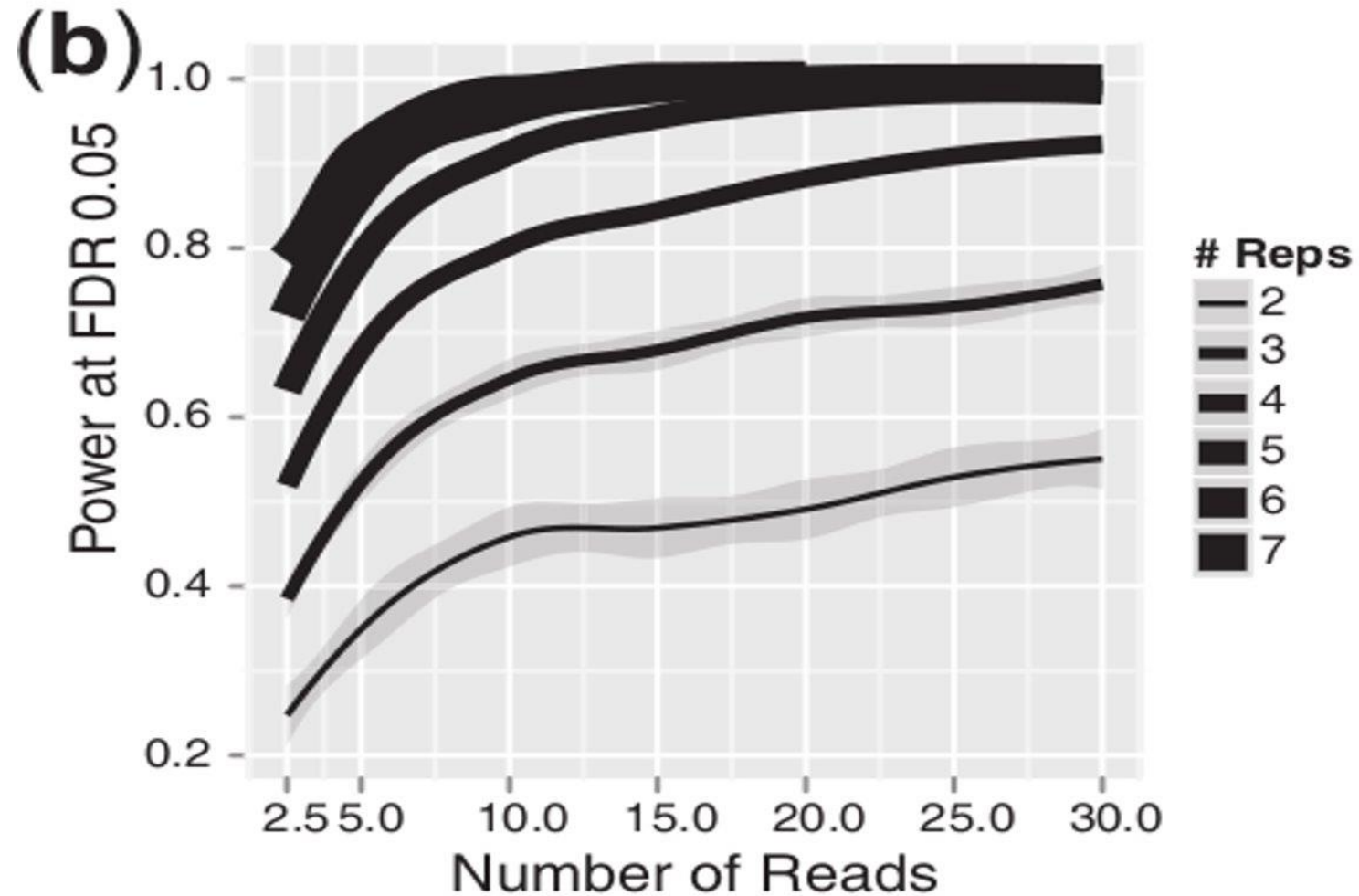


2

5

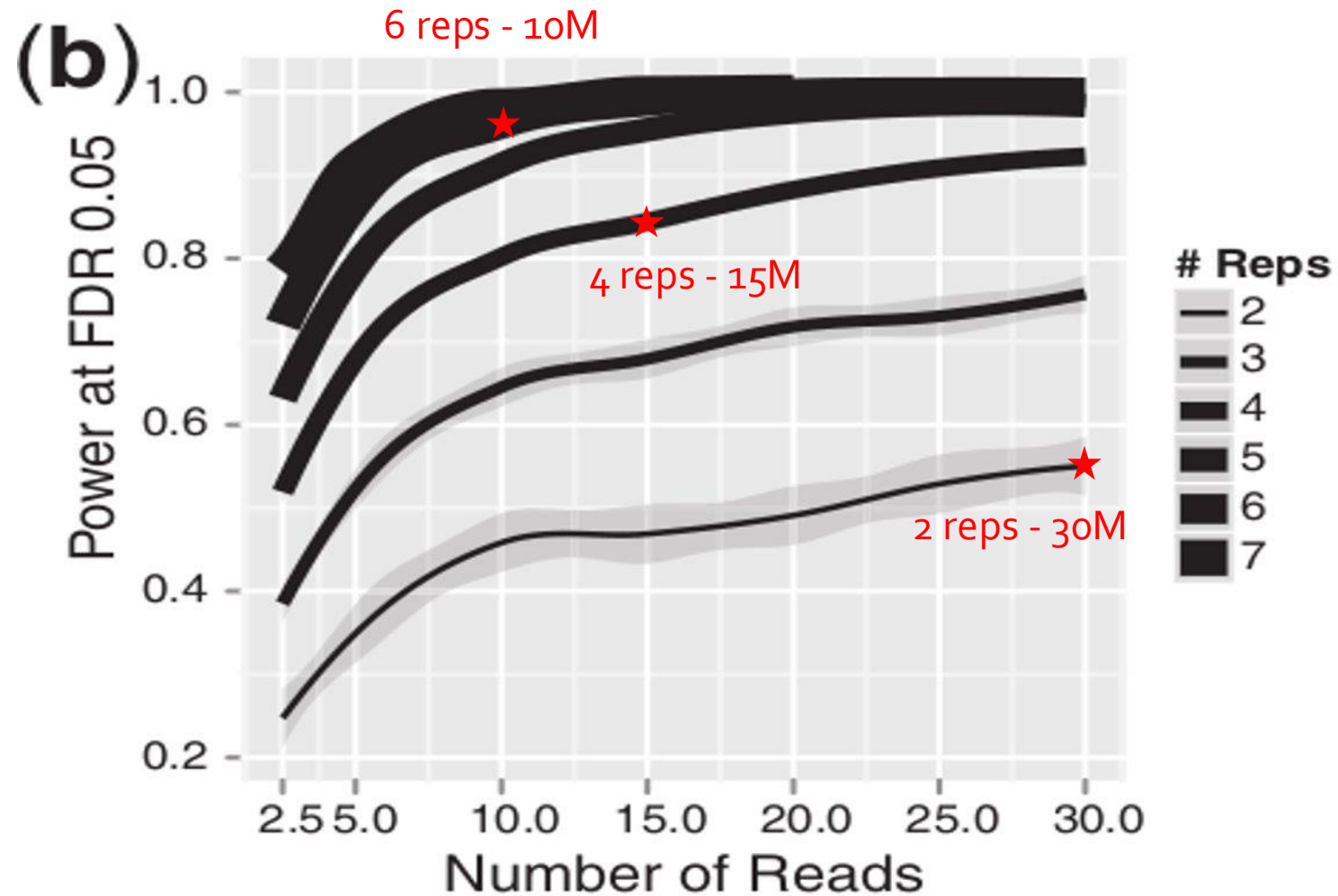
10

More reads or more replicates?



From Liu et al. 2014. RNA-seq differential expression studies: more sequence or more replication?

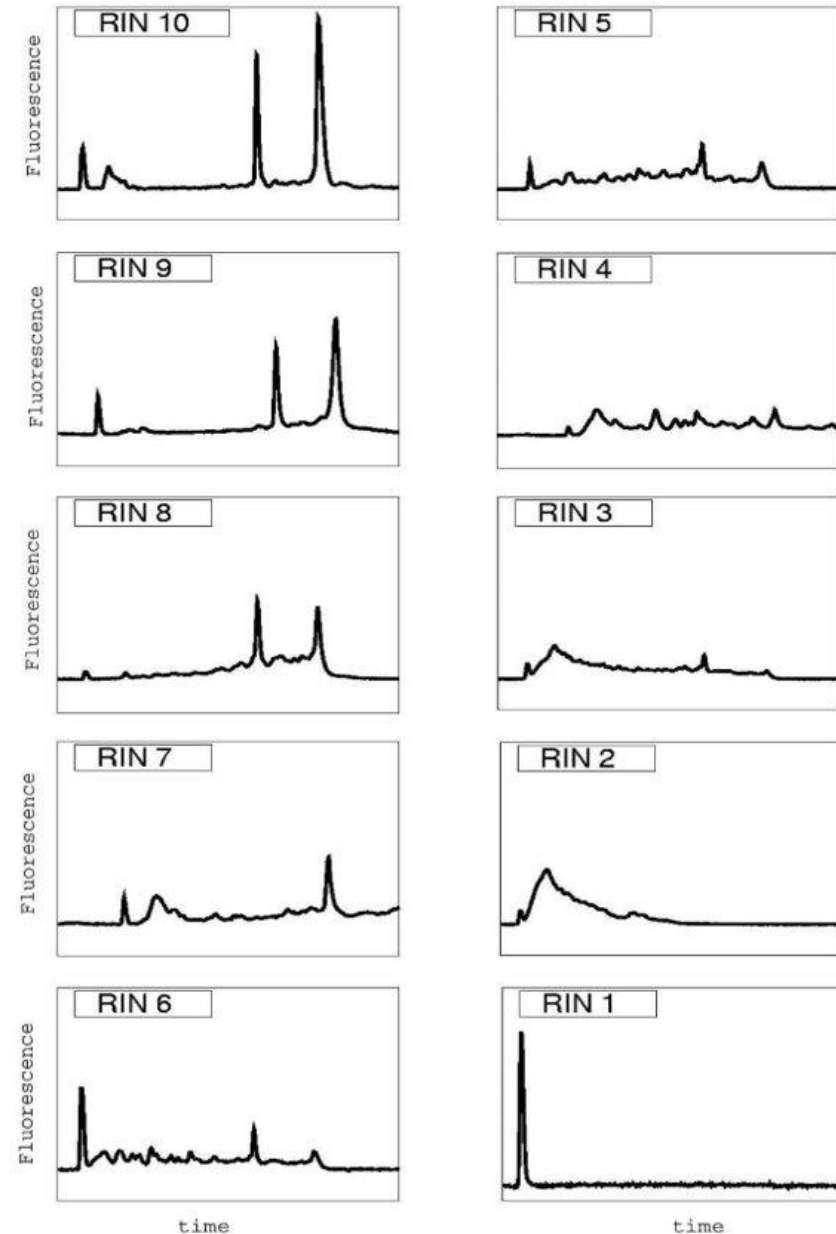
More reads or more replicates?



From Liu et al. 2014. RNA-seq differential expression studies: more sequence or more replication?

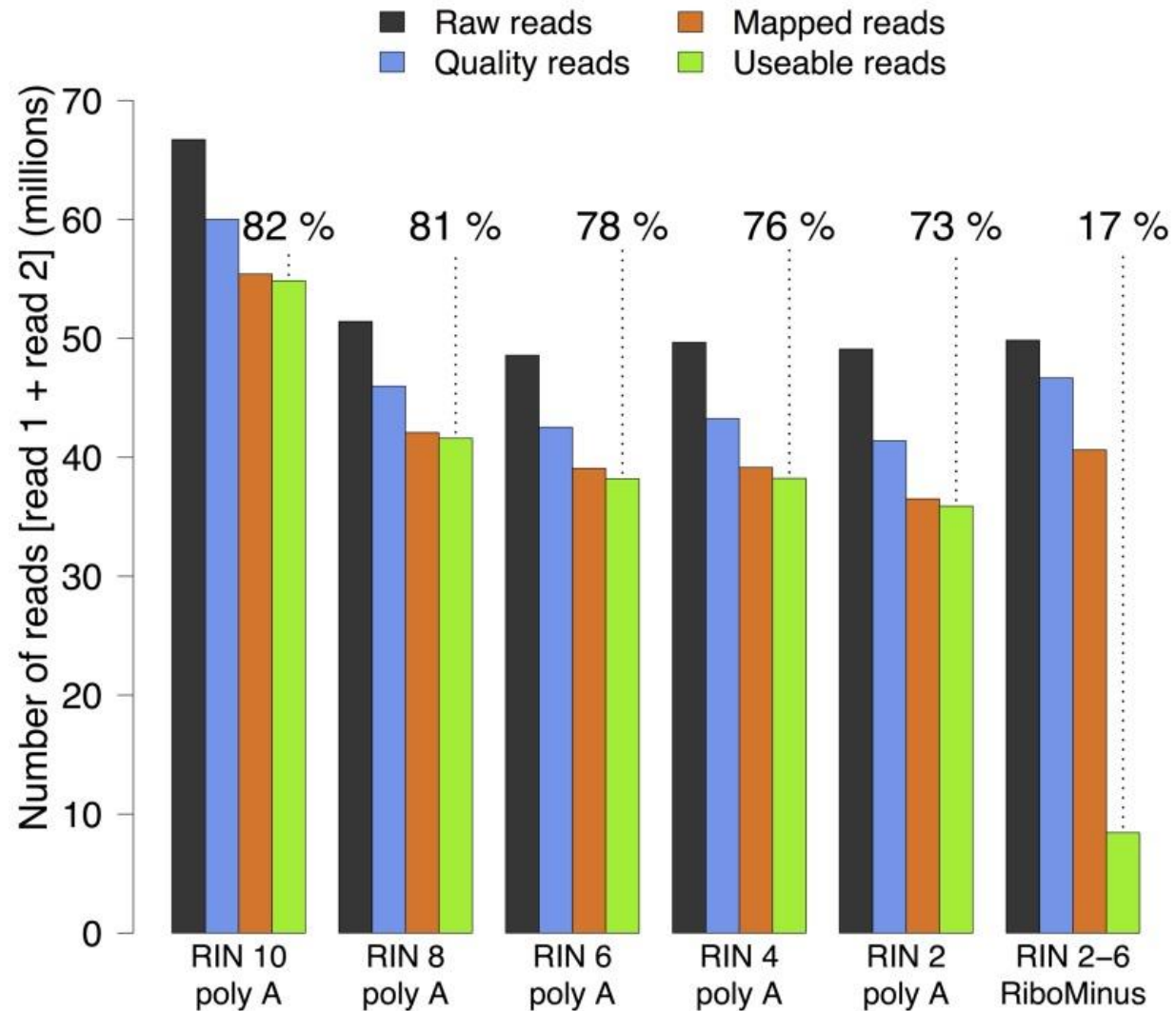
RNA sample preparation - RIN

- Sample quality is critically important: we cannot make up for poor data
- RNA Integrity Number (RIN)
- Minimums:
 - 7-8 : eukaryot mRNA
 - 9 : bacterial

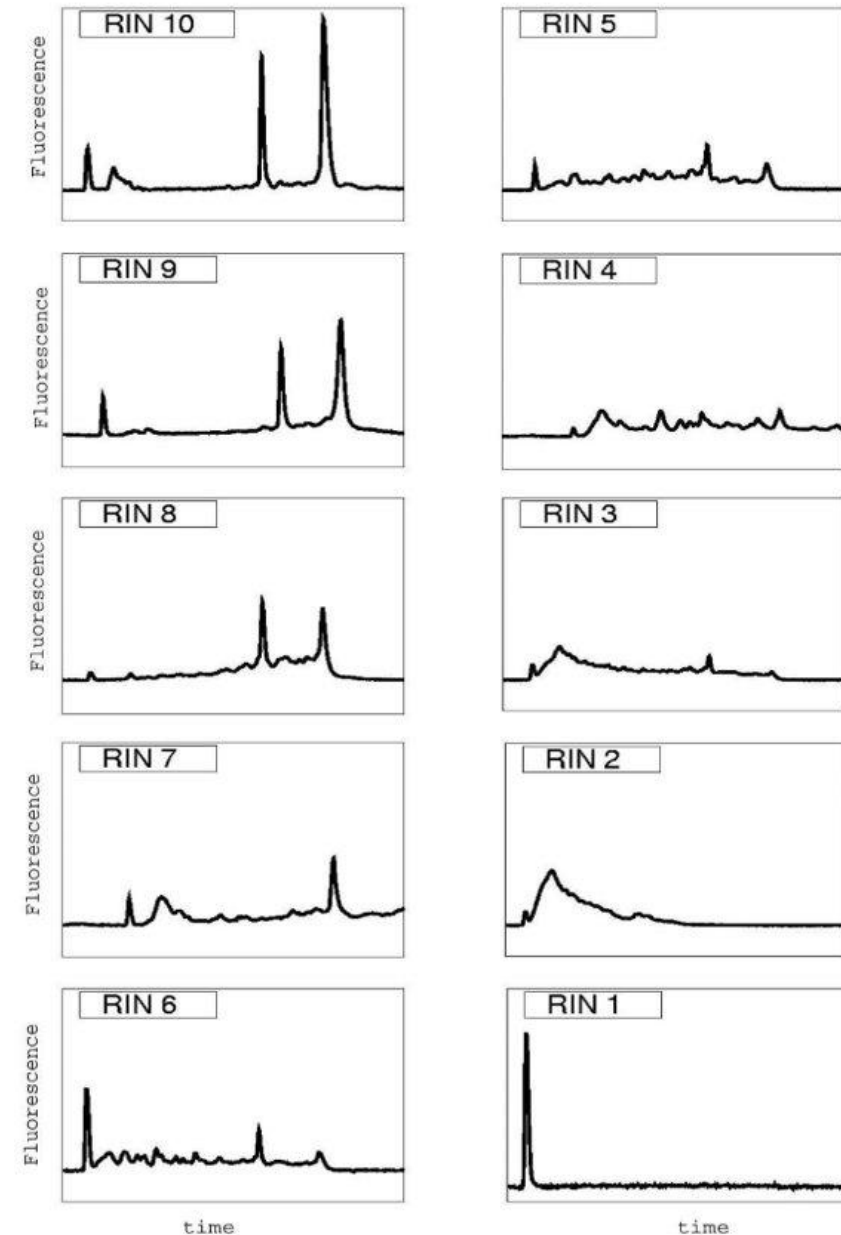


RNA sample preparation - RIN

Effects of preprocessing analysis pipeline



Schroeder *et al* BMC Mol Biol 2006



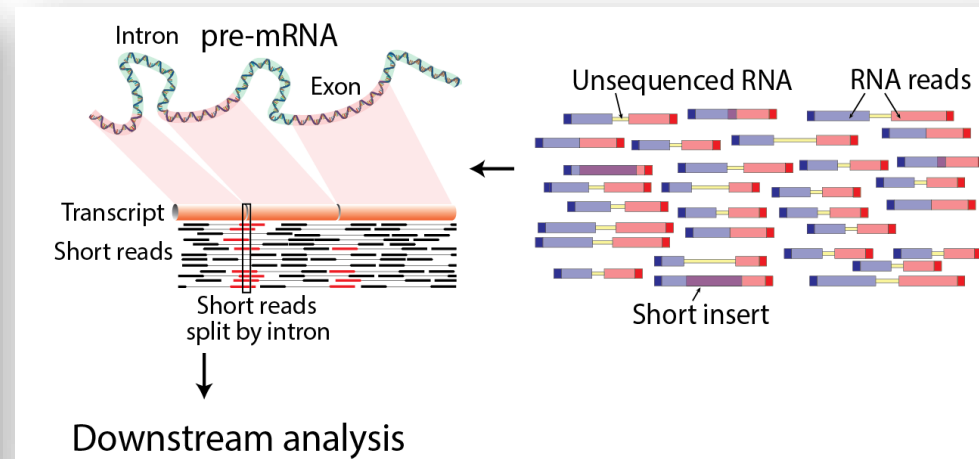
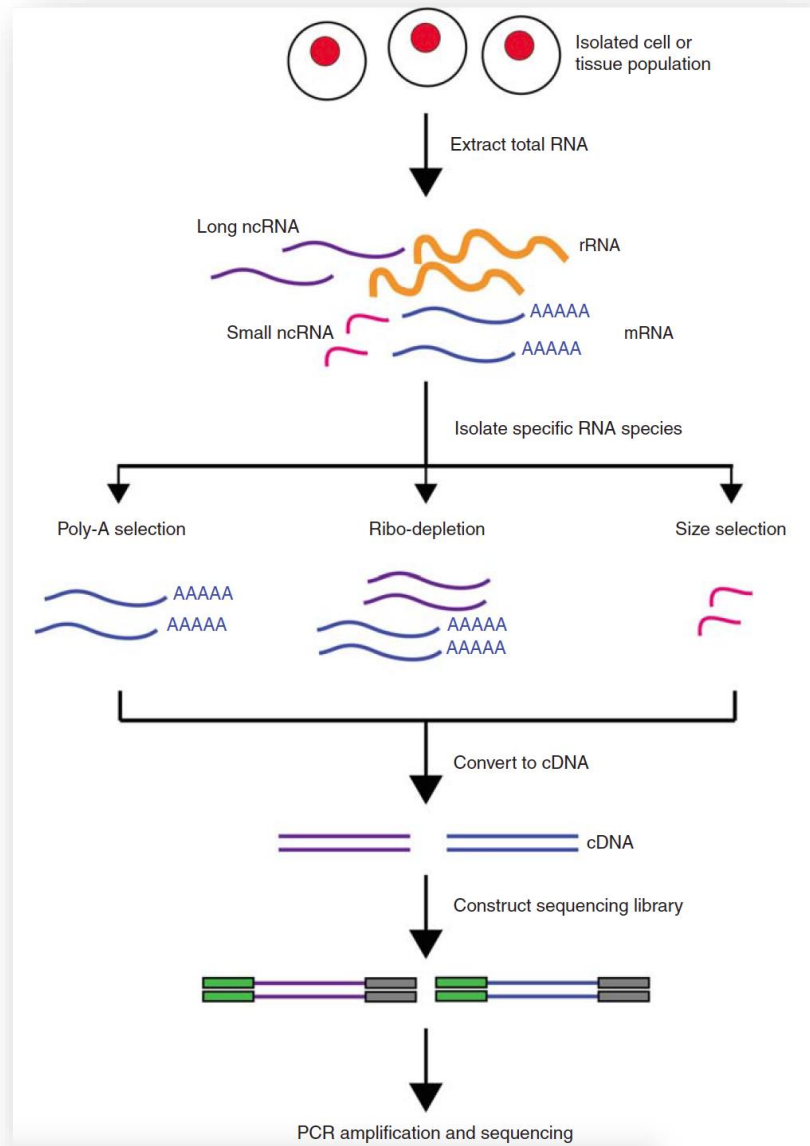
Sigurgeirsson B, Emanuelsson O, Lundeberg J. Sequencing degraded RNA addressed by 3' tag counting.

PLoS One. 2014 Mar 14;9(3):e91851.

slides Outline

- RNA and molecular biology
- Main challenges for RNAseq
- Major Sequencing technologies
- Planning your sequencing : choices, number of samples, ...
- **Bioinformatics analysis overview**

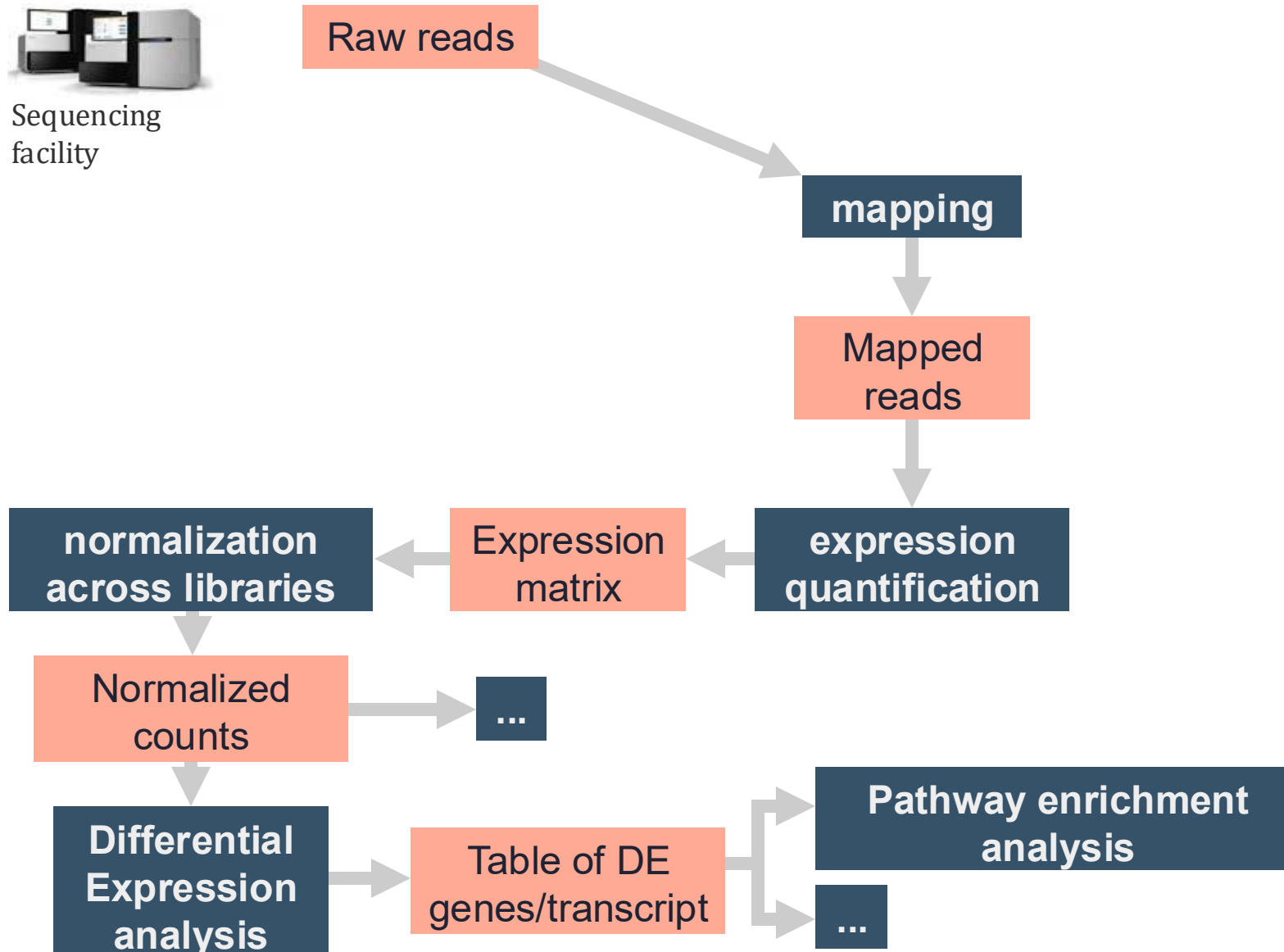
Basic RNAseq protocol overview



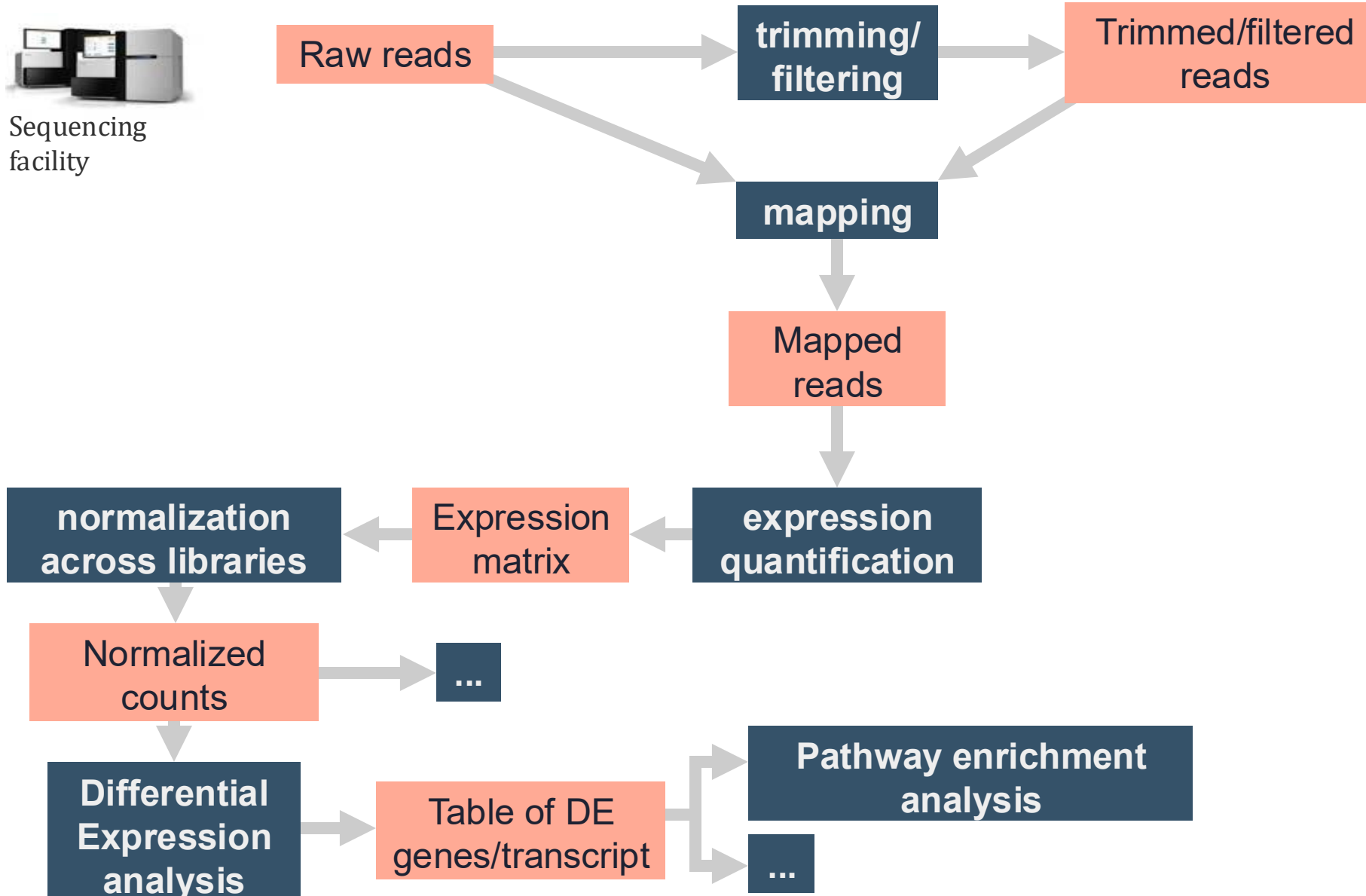
RNAseq data analysis - basic pipeline



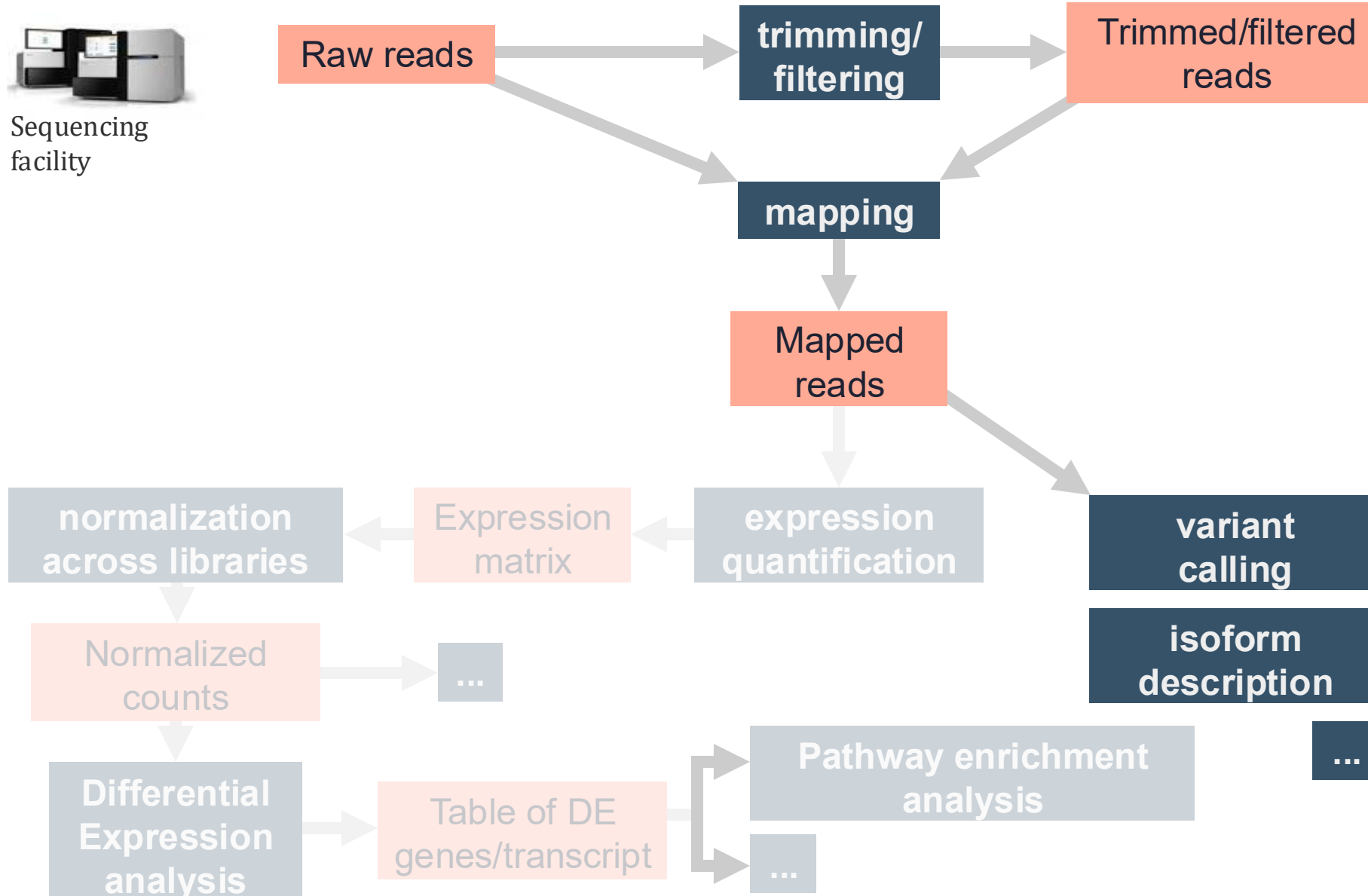
Sequencing
facility



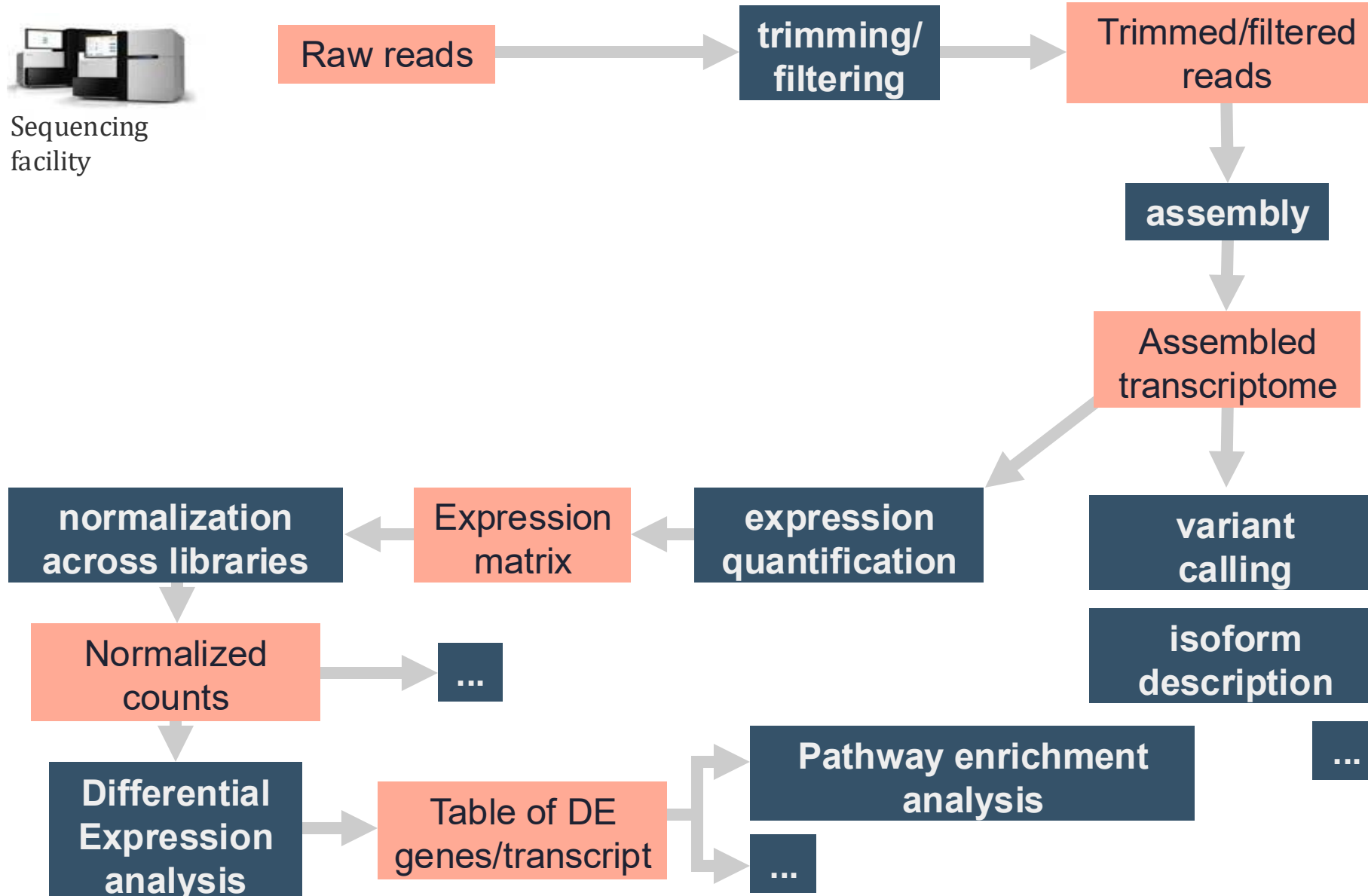
RNAseq data analysis - basic pipeline



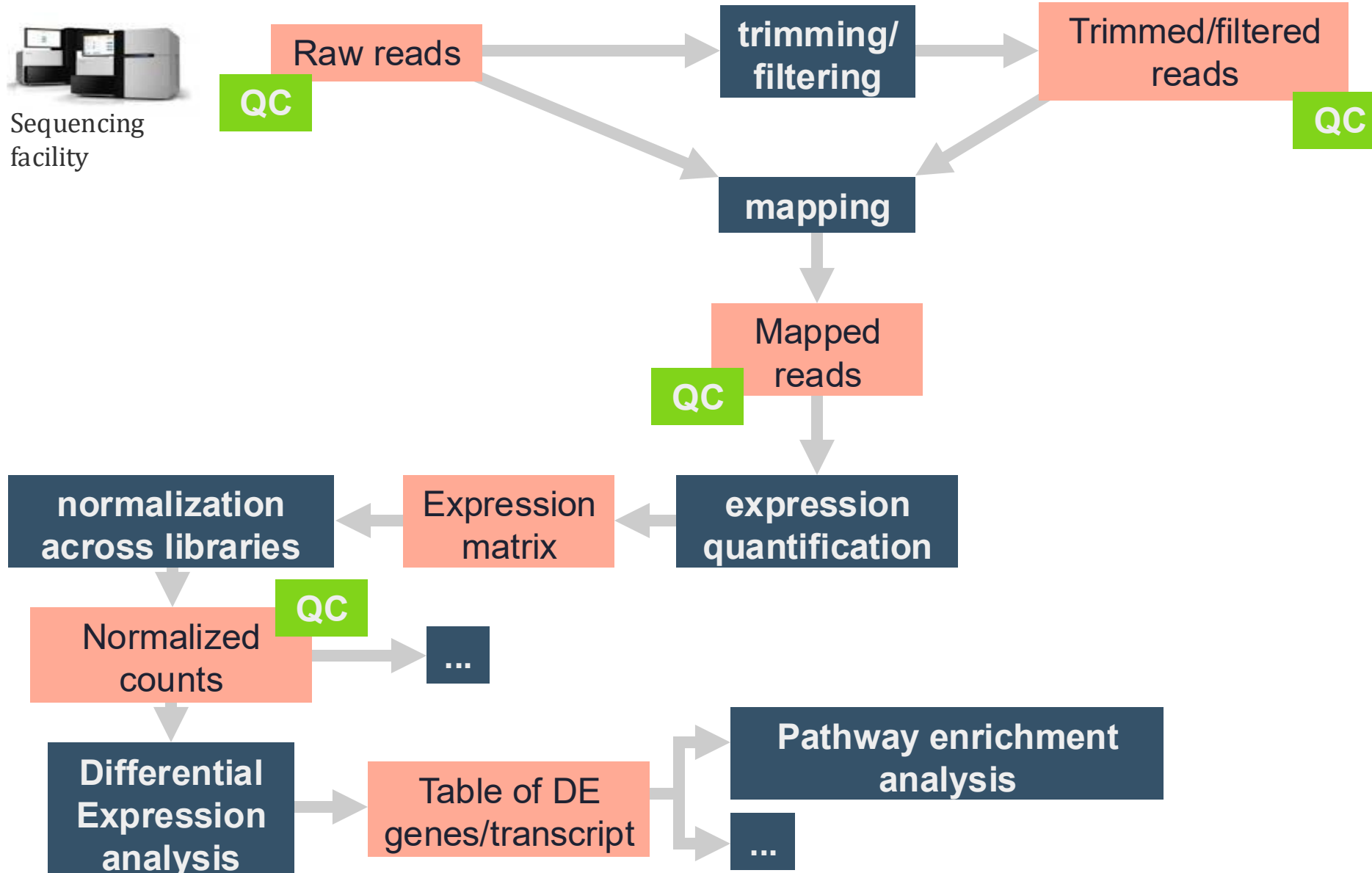
RNAseq data analysis - basic pipeline

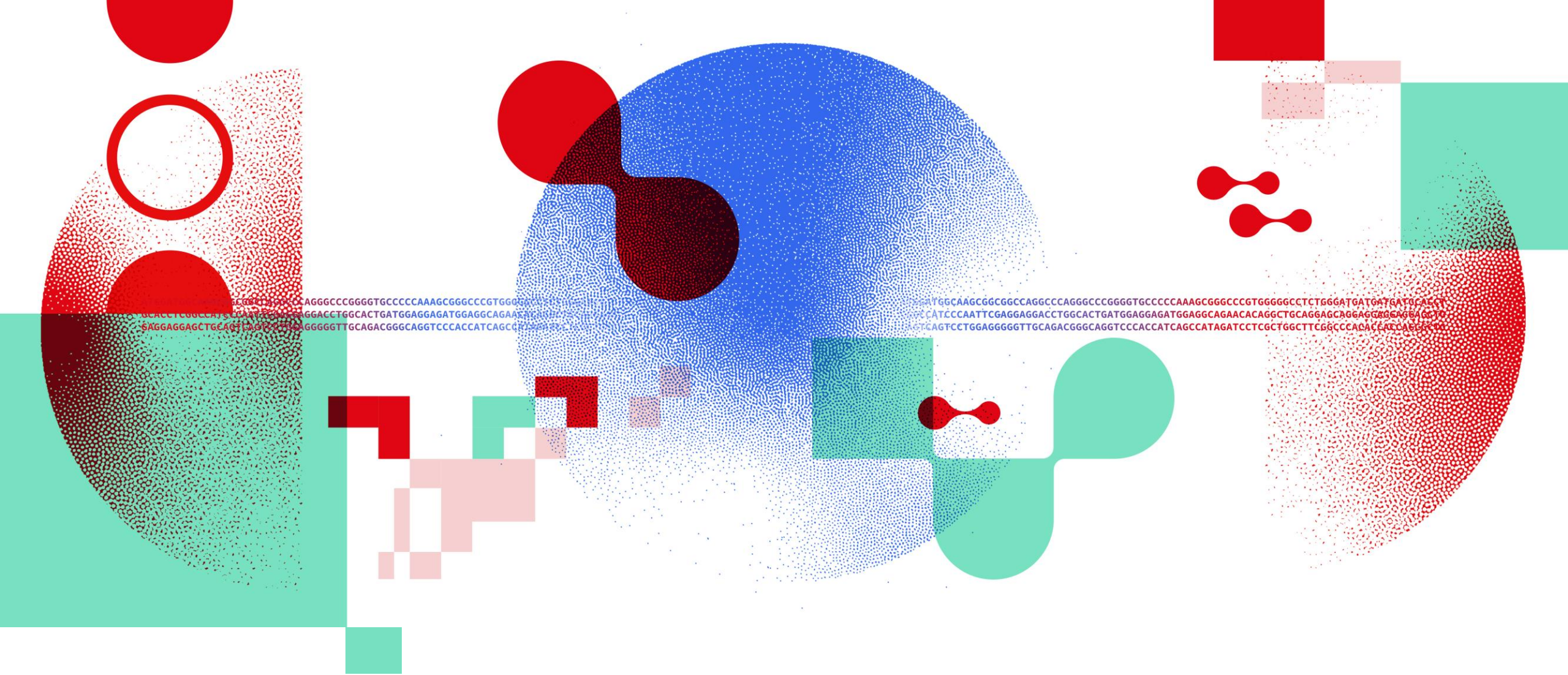


RNAseq data analysis - basic pipeline



RNAseq data analysis - basic pipeline





Thank you